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Quality-Driven Next-Best-View Planning for Target-Conditioned Egocentric 3D Reconstruction

Qualitätsgetriebene Next-Best-View-Planung für zielkonditionierte egozentrische 3D-Rekonstruktion

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Abstract

This thesis studies target-conditioned, quality-driven Next-Best-View (NBV) planning for egocentric 3D reconstruction in Aria Synthetic Environments (ASE). It defines target-specific Relative Reconstruction Improvement (RRI) as an oracle signal, constructs finite candidate and replay contracts, and separates privileged label generation from the actor-visible inputs available to a learned finite-horizon value model. The evaluation is designed to measure oracle-lookahead headroom and the fraction recovered by a learned policy under matched oracle re-evaluation. The evidence available for this version establishes the evaluation contract and implementation readiness, but does not contain confirmatory held-out policy outcomes; it therefore supports no claim that a learned policy improves over the specified baselines. The main remaining limitation is the absence of a validated, population-level rollout bundle with paired endpoint estimates and uncertainty.

Zusammenfassung

Diese Arbeit untersucht zielkonditionierte, qualitätsgetriebene Planung der nächsten besten Ansicht für die egozentrische 3D-Rekonstruktion in ASE. Sie definiert die zielspezifische RRI als Orakelsignal, legt Verträge für endliche Kandidatenmengen und Replay-Daten fest und trennt die privilegierte Erzeugung von Trainingssignalen von den für ein gelerntes Modell mit endlichem Horizont sichtbaren Eingaben. Die Evaluation soll den Spielraum einer vorausschauenden Orakelstrategie und den durch eine gelernte Strategie erreichten Anteil unter identischer Orakel-Neubewertung messen. Die für diese Fassung verfügbare Evidenz belegt den Evaluationsvertrag und die Implementierungsbereitschaft, enthält jedoch keine bestätigenden Ergebnisse auf zurückgehaltenen Daten. Daher wird keine Überlegenheit einer gelernten Strategie gegenüber den festgelegten Baselines behauptet. Die wesentliche verbleibende Einschränkung ist das Fehlen eines validierten Rollout-Datensatzes auf Populationsebene mit gepaarten Endpunktschätzungen und Unsicherheitsangaben.

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Glossary and Abbreviations

Dataset

ADT – Aria Digital Twin: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

AEO – Aria Everyday Objects: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

ASE – Aria Synthetic Environments: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

CPF – Central Pupil Frame: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

MPS – Machine Perception Services: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

MTD – Motion Trajectory Data: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

MFCD – Multi-Frame Camera Data: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the

egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

MSDPD – Multi-Semi-Dense Point Data: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

Project Aria – Project Aria: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

SSL – SceneScript Language: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

snippet – Snippet: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

VRS – Virtual Reality Standard: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

target – Target of Interest: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom:** Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum:** Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward:** Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box:** 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom:** Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve:** Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance:** Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \rightarrow M}$ and $D_{M \rightarrow P}$; older seminar material may still call this CD.

CR – Coverage Ratio: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

GT – Ground Truth: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

RRI – Relative Reconstruction Improvement: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

reward – Target Root-Gain Reward: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

target RRI – Target-Specific RRI: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

Q-CORAL – CORAL Q-Value Decoder: Ordinal finite-horizon value decoder that maps continuous fitted-Q targets to fixed cumulative classes and decodes repaired class mass through fixed continuous-Q representatives.

EFM3D – Egocentric Foundation Model 3D: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

EVL – Egocentric Voxel Lifting: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

Q_H – Finite-Horizon Q Function: Target-conditioned finite-candidate value family $Q_{\theta}(s,e,a,h)$ that scores each valid candidate for an explicit requested residual horizon h up to a configured maximum H . The thesis-core architecture is one shared horizon-conditioned scorer; dense Q_1 and exact Q_2 targets establish the recursion, horizon-indexed fitted Q supplies longer-horizon targets, and Double Q remains an optional correction for a noisy learned successor maximum.

target-conditioned scorer – Target-Conditioned Scorer: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

VIN – View Introspection Network: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

cost – Acquisition Cost: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

candidate – Candidate View: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

transition – Counterfactual Transition: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

action set – Finite Candidate Action Set: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

return – Finite-Horizon Return: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

5DoF – Five Degrees of Freedom: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

CF+ state – Geometry-Rich Counterfactual State: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic

observations only: rendered depth, valid mask, backprojected point cloud, and derived local normals or support summaries.

historic state – Historic Snippet State: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

CF0 state – Minimal Counterfactual Actor State: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

NBV – Next-Best-View: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

oracle state – Oracle Rollout State: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

offline state – Persisted Offline Sample State: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

state – Rollout State: Family of rollout state variants used by ARIA-NBV.

The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

NBV MDP – Target-Conditioned NBV MDP: Finite-horizon Markov decision process used to define ARIA-NBV’s target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

mask – Validity Mask: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. The scalar mask $m_{t,i}$ gates selectable rows, while the reason code $\rho_{t,i}$ records why a row was rejected. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

GT-EVAL – Ground-Truth Target Evaluation: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

OBS-SEL – Observed Target Selection: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

PRED-Q – Predicted-Target Q: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target

descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

MVS – Multi-view Stereo: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV's current oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

PC – Point Cloud: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

SLAM – Simultaneous Localization and Mapping: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

track – Track: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

VIO – Visual-Inertial Odometry: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

Occupancy Grid – Occupancy Grid: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL

outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

3DGS – 3D Gaussian Splatting: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

***oracle RRI* – Oracle RRI:** RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
Oracle and Reconstruction		
RRI	Relative Reconstruction Improvement score for a candidate view.	oracle.rri
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	oracle.points
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	oracle.points_q
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	oracle.candidates
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	oracle.candidates_t
$q_{t,i}$	Candidate pose i at rollout step t.	oracle.candidate_qti
$\mathbf{c}$	World-space camera-center vector; subscripts identify a rollout root or candidate pose.	oracle.center
D_q	Oracle-rendered or candidate-specific depth map for a proposed view.	oracle.depth_q
$D_{P \rightarrow M}$	Point-to-mesh directional error component.	oracle.acc
$D_{M \rightarrow P}$	Mesh-to-point directional error component.	oracle.comp
D	Aggregate point-mesh reconstruction error used by RRI definitions.	oracle.err
ASE Assets		
$\mathcal{M}^{\text{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	ase.mesh
$\mathcal{M}_e^{\text{GT}}$	Ground-truth mesh crop for target e.	ase.mesh_target
$\mathcal{F}^{\text{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	ase.faces
$T_{\text{rig}}^{\text{w}}(t)$	Project Aria rig pose in the world frame at time t.	ase.traj
$T_{\text{rig}}^{\text{w}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	ase.traj_final
VIN Scorer		
r	Target or scene RRI value used as a model target.	vin.rri
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	vin.rri_hat
$\mathcal{L}$	Training loss for scorer or value-model objectives.	vin.loss
m	Candidate validity indicator used by scorer diagnostics.	vin.cand_valid
Entity and Target		

Symbol	Meaning	Key
RRI_e	Target-specific Relative Reconstruction Improvement for entity e .	<code>entity.rri_e</code>
$\text{RRI}_{t,i}^e$	State-relative marginal target RRI for candidate i at rollout step t .	<code>entity.target_rri_marginal</code>
$C_t^{\text{RRI},e}$	Running sum of selected state-relative target RRIs.	<code>entity.target_rri_cumulative</code>
J_t^e	Running cumulative target gain normalized by rollout-root error.	<code>entity.target_root_gain_cumulative</code>
φ_e	Actor-visible target/entity descriptor used to condition target-specific value prediction.	<code>entity.target_desc</code>
p_e^w	World-space center of the selected target entity e .	<code>entity.center</code>
Planning and Rollout		
s	Abstract planning state.	<code>rl.s</code>
a	Action index selecting a candidate row.	<code>rl.a</code>
r	Scalar reward or immediate gain.	<code>rl.r</code>
G	Finite-horizon return accumulated across rollout steps.	<code>rl.G</code>
$\mathcal{M}_{\text{NBV}}$	Target-conditioned finite-candidate NBV decision process.	<code>rl.mdp_nbv</code>
$\mathcal{A}(s_t)$	Valid action set available in state s_t .	<code>rl.action_set</code>
T	State-transition operator for selected candidate actions.	<code>rl.transition</code>
s_t^{hist}	Logged historic snippet state at rollout step t .	<code>rl.s_hist</code>
s_t^{off}	Persisted offline sample state used by training readers.	<code>rl.s_off</code>
s_t^{cf0}	Minimal actor-visible counterfactual state.	<code>rl.s_cf0</code>
$s_t^{\text{S0-pose}}$	Implemented fixed-root pose-history state used by <code>qh_cf0_v1</code> .	<code>rl.s_pose</code>
$s_t^{\text{cf+}}$	Geometry-enriched counterfactual successor state.	<code>rl.s_cf_geom</code>
$s_t^{\text{CF-GT-carrier}}$	Implemented privileged selected-depth carrier used by <code>qh_cfplus_gt_depth_v1</code> .	<code>rl.s_cf_gt_carrier</code>
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	<code>rl.s_oracle</code>
r_t^e	Target-specific reward at rollout step t .	<code>rl.reward_target</code>
Entity and Target		
r_t^e	Canonical target-specific reward at rollout step t .	<code>entity.target_reward</code>
Planning and Rollout		
$G_t^{(H)}$	H-step finite-horizon return from rollout step t .	<code>rl.return_h</code>
Q_H	Finite-horizon candidate-value function.	<code>rl.qh</code>
$Q_{h,\theta,e,i}^{\text{cond}}$	Action-mask-independent conditional candidate value emitted by the scorer.	<code>rl.conditional_q</code>

Symbol	Meaning	Key
$\ell_{t,i}^{\text{feas}}$	Physical or observed feasibility logit emitted by the scorer's feasibility head.	rl.feasibility_logits
e_k^Q	Fixed continuous fitted-Q boundary between adjacent CORAL classes.	rl.coral_q_edge
u_k^Q	Fixed continuous-Q representative used to decode one CORAL class.	rl.coral_q_value
c_n^Q	Ordinal class assigned to one continuous fitted-Q target.	rl.coral_q_label
γ	Return discount factor.	rl.gamma
H	Planning or rollout horizon length.	rl.H
$H_{\max}$	Maximum residual horizon admitted by a scorer/data contract.	rl.H_max
h	Scalar residual horizon requested for one scorer query; it must satisfy $1 \leq h \leq b_t \leq H_{\max}$.	rl.requested_horizon
$m_{t,i}$	Hard validity mask for candidate i at rollout step t.	rl.validity_mask
$m_{t,i}^{\text{cand}}$	Materialized candidate row versus structural padding.	rl.candidate_row_mask
$m_{t,i}^{\text{act}}$	Authoritative physically valid action support.	rl.action_mask
$m_{t,i}^{Q,h}$	Availability of a finite factual value target for candidate i at requested horizon h.	rl.q_label_mask
$m_{t,i}^F$	Availability of a trustworthy feasibility label for candidate i.	rl.feasibility_label_mask
m_t^{succ}	Availability of a factual successor backup for transition t.	rl.successor_mask
$\ell_{t,i}^{\text{src}}$	Actor-oracle source-provenance role for candidate evidence.	rl.source_role
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t.	rl.invalid_reason
e_t	Selected target entity at rollout step t.	rl.target
b_t	Remaining acquisition budget at rollout step t.	rl.budget
$C(\tau)$	Acquisition cost of a selected trajectory.	rl.acquisition_cost
Shape and Size		
N_q	Number of candidate views in a candidate table.	shape.Nq
K	Number of ordinal bins or discrete levels when used in scorer losses.	shape.K
scene		
M_t^{ray}	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t.	scene.ray_memory_t
Observation and Actor State		
$F_t^{\text{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	obs.dino_point_bank_t

Symbol	Meaning	Key
$\mathbf{X}_t^{\text{pt}}$	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	obs.point_tokens_t
scene		
Φ_t^{scene}	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	scene.scene_memory_t
$E_0^{\text{EVL-local}}$	Root local EVL evidence field used for target/candidate support reads.	scene.evl_local
$\omega_{t,i}^{\text{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	scene.evl_support_frac
$g_{t,i}^{\text{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	scene.evl_support_token
g_e^{tgt}	Pooled target-support descriptor for the selected target hypothesis.	scene.target_support_pool
$g_{t,i}^{\text{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	scene.frustum_support_pool
$g_{t,e,i}^{\text{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	scene.target_frustum_pool
$g_{t,i}^{\text{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	scene.ray_query_ti
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	scene.render_query
spatial		
r_t	Reference pose for candidate-relative descriptor construction at decision step t.	spatial.ref_pose
$T_{r_t,i}^{\text{rel}}$	Relative transform from the reference pose r_t to candidate camera i.	spatial.ref_candidate_transform
$h_{t,i}^{\text{pose}}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	spatial.candidate_pose_feat
$h_{t,e i}^{\text{rel}}$	Candidate-target relation descriptor in the candidate/query local frame.	spatial.candidate_target_rel_feat
$e_{a i}^{\text{rel}}$	Query-local relative positional embedding for target, history, support, or candidate relations.	spatial.relation_rpe
model		
h_e^{tgt}	Learned selected-target token built from the actor-visible target descriptor and scene support.	model.target_token

Symbol	Meaning	Key
$\mathbf{x}_{t,i}$	Per-candidate row feature assembled from pose, relation, support, validity, provenance, and history descriptors.	model.candidate_row
Entity and Target		
$J_e^{(H)}$	Endpoint reconstruction gain for target e over horizon H.	entity.endpoint_gain
$J_{e,\log}^{(H)}$	Log-scale endpoint reconstruction gain for target e.	entity.log_gain
Δ_{look}	Gain available to an oracle lookahead policy beyond one-step selection.	entity.lookahead_headroom
η_Q	Fraction of oracle lookahead gain recovered by the Q policy.	entity.q_recovery
$G_t^{(H)}$	Target-conditioned finite-horizon return.	entity.return_h
Observation and Actor State		
$\mathbf{D}$	Depth observation sequence.	obs.depth
$\mathbf{n}$	Per-point or per-face normal observation.	obs.face_normal
$\mathbf{I}^{\text{rgb}}$	RGB image observation.	obs.img_rgb
$\mathcal{P}^{\text{cf}}$	Counterfactual point observation.	obs.points_cf
$\mathcal{P}^{\text{semi}}$	Semi-dense observed point set.	obs.points_semi
$\mathcal{P}_t$	Point set available at rollout step t.	obs.points_t
Planning and Rollout		
$\mathcal{Q}_t$	Finite candidate-view table at rollout step t.	rl.candidate_table
VIN Scorer		
$\mathbf{F}_v$	Learned value field evaluated at v.	vin.field_v

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1 Introduction

Active reconstruction couples sensing and inference: after a fixed acquisition budget, reconstruction quality depends not only on the surface estimator but also on which feasible viewpoints were acquired. Classical active perception therefore treats sensing actions as part of perception, and view-planning work formalizes the recurring generate–score–select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. This thesis studies a deliberately bounded instance of that problem: target-conditioned view selection from a finite admissible candidate table for egocentric indoor reconstruction.

VIN-NBV provides the closest objective precedent by supervising candidate ranking with RRI, computed from the reduction in point–mesh reconstruction error after adding a query view [Fra+25]. ARIA-NBV retains this quality-driven principle but changes the task and planning question. The utility is target-specific rather than scene-wide, and non-myopic value is interpreted only if bounded oracle lookahead improves endpoint quality over one-step selection under the same finite candidate support. Project Aria, ASE, and Egocentric Foundation Model 3D (EFM3D) provide the calibrated egocentric, mesh-supervised, and local actor-visible substrate for this study [Eng+23, Met25a, Str+24].

The current data-generation tasks are defined directly from geometry-valid GT oriented bounding boxes. They provide target identity, target crops, and oracle labels, but they do not implement actor-visible target discovery or identity matching. Consequently, deployable-input claims require a separate observed- or predicted-target protocol; the current target tasks support oracle supervision and controlled evaluation only.

1.1 Objectives and boundary

The thesis objective is a finite-candidate, target-conditioned NBV comparison with equal acquisition budgets and oracle re-evaluation. The boundary is explicit: GT geometry and meshes provide labels and evaluation only; the actor-visible policy receives observed or predicted target descriptors and validity masks. Claims

are restricted to the sampled candidate support, horizon, branch factor, target protocol, and held-out split.

Problem statement. Given a GT-defined target task, actor-visible reconstruction evidence, a finite candidate table with hard validity constraints, and a fixed acquisition budget, determine whether bounded oracle lookahead improves endpoint target reconstruction over one-step oracle-greedy selection and, only if such headroom exists, whether a learned policy using non-privileged inputs recovers a meaningful fraction of it under the same oracle evaluation.

The thesis separates supervision from decision-time information. Ground-truth geometry defines current target tasks, target crops, oracle labels, and evaluation; it is not an ordinary actor input. Invalid candidates are outside the admissible action set rather than examples with low utility. One-step oracle greedy is an immediate-reward comparator over the evaluated valid candidate table, while oracle lookahead is an upper reference only within the fixed candidate generator, horizon, branch factor, target pool, and validity regime. These restrictions make negative results interpretable without turning a bounded experiment into a universal statement about view planning.

1.2 Research Questions

The thesis asks whether ARIA-NBV can perform target-conditioned, quality-driven multi-step NBV (NBV) selection with a finite candidate action space. The utility is target-specific RRI (RRI), and every learned action is re-evaluated with the oracle under the same candidate support, validity rules, and acquisition budget. GT (GT) geometry defines target tasks, labels, and evaluation; it is not an ordinary actor input.

1.2.1 Objectives and scope

The work has six linked objectives. First, validate the geometry, candidate-label, invalidity, split, and oracle-RRI contracts on a trusted ASE subset before scale-up. Second, define the observed-target-selection / predicted-target- Q / GT-evaluation protocol. Third, establish a learned one-step target-conditioned scorer as the

required myopic control. Fourth, validate mixed candidate sets and rollout support before training a finite-horizon value model. Fifth, train $Q_{H,\theta}$ from ASE oracle rollouts as a finite-candidate value model, first using candidate-to-state queries. Finally, grow support from trusted subsets to scene-level ASE evidence only while preserving mesh-supervised target RRI.

Online discrete Q_H is a deferred RQ5 bridge; continuous or simulator-backed actor–critic control is RQ6 and is not required for the finite-candidate thesis result. The active quantitative core is therefore RQ1–RQ4 below.

Development provenance and reconciliation. The reviewed Quarto source (the reviewed historical QMD at `origin/main`) supplied the six-tier ordering: RQ1 objective, RQ2 offline finite-candidate planning, RQ3 actor-visible target representation, RQ4 candidate/rollout/scale support, RQ5 online discrete Q_H , and RQ6 continuous or simulator escalation. The base Typst formulation on this branch had consolidated the first four questions but omitted the two explicitly gated extensions. The current formulation preserves the reviewed ordering, makes RQ1–RQ4 the evaluated core, and states RQ5/RQ6 as conditional bridges. The Quarto page remains historical review input; this section and the executable Python/configuration owners define current terminology and implementation status.

Implementation: planned Evidence: pending

The actor-visible descriptor and deterministic 3D-IoU matching below are a planned hypothesis and acceptance protocol, not an implemented result.

Promotion gate: Observed/predicted target matching and target-conditioned scoring remain future work; the current oracle-task protocol is the available baseline

1.2.2 RQ1 — Objective and endpoint contract

Can target-conditioned, finite-candidate NBV improve endpoint quality for a selected target under a fixed acquisition budget while keeping training return, endpoint evaluation, and acquisition cost separate? For a rollout rooted at s_0 with target e , let $\mathcal{P}_t$ be the accumulated fused points and $\mathcal{M}_e^{\text{GT}}$ the matched target surface. The oracle-only target crop gives

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The primary fixed-budget metric is endpoint gain J_H^e . The default multi-step reward is root-normalized target gain,

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

State-relative target RRI remains a diagnostic and VIN-compatible one-step label. We report both its marginal and selected-chain cumulative forms,

$$\text{RRI}_{t,i}^e = \frac{\Delta_t^e - \Delta_{t|i}^e}{\max(\Delta_t^e, \varepsilon)}$$

$$C_t^{\text{RRI},e} = \sum_{k=0}^{t-1} \text{RRI}_{k,a_k}^e$$

without treating their non-telescoping sum as endpoint gain. Log-error gain and motion or feasibility costs are explicit ablations. Invalid candidates are hard constraints, never the lowest-RRI class. Report endpoint gain, cumulative root gain, diagnostic target RRI, scene RRI, view count, path length, invalid-action rate, and runtime at matched budgets.

1.2.3 RQ2 — Offline finite-candidate planning

Does bounded oracle lookahead have positive endpoint-quality headroom over one-step oracle-greedy selection, and can an offline $Q_{H,\theta}$ policy recover a measurable fraction of it? The first comparison fixes target tasks, roots, candidate support, validity, horizon, and budget. Define

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

If this headroom is indistinguishable from zero for the evaluated support, the result is a setup-specific negative finding, not a claim that target RRI is universally myopic. When it is positive, $Q_{H,\theta}$ is trained from ASE oracle rollout traces and

compared with random-valid, one-step oracle greedy, learned one-step scoring, bounded oracle lookahead, and oracle-scored temperature-softmax traces. The learned model emits one masked bounded-horizon value per finite candidate; GT meshes, crops, and oracle values remain supervision/evaluation only. Gumbel-Top- k is later diversity evidence, not a prerequisite for the first run.

1.2.4 RQ3 — Actor-visible target representation

Which actor-visible target descriptor and matching protocol support one-step and finite-horizon scoring without privileged target geometry or all-candidate oracle renders at decision time? The proposed protocol separates observed or predicted target selection, predicted-target conditioning, and ground-truth target-crop evaluation. The first descriptor contains observed/predicted Oriented Bounding Box (OBB) centre, extents, orientation, class, confidence, projected area, relative pose, semi-dense point support, and EVL support. A compact crop descriptor from actor-visible spatial evidence is the first ablation; entity tokens and appearance features are later ablations. A ground-truth target box is retained only as a privileged sanity or upper-bound path, not as an actor-visible deployment input.

The planned matcher first requires compatible semantic class, then applies deterministic 3D IoU with an explicit threshold τ_{IoU} . If the best and second-best compatible matches are separated by less than the explicit ambiguity margin τ_{gap} , the proposal is rejected as ambiguous; below-threshold matches are unmatched. Visibility, projected area, and semi-dense/EVL support are audit-only evidence fields for this association rule, not association scores. Unmatched or ambiguous targets are protocol-invalid cases, not low-RRR examples.

1.2.5 RQ4 — Candidate, rollout, and scale support

Do mixed target-centric and default-exploration candidates, controlled rollout branching, and scale-aware replay evidence improve reliability over a narrower candidate family? The thesis-core finite table is the executable 60-row mixture: forward-local (24), target-bearing-local (24), and lateral-target-bypass (12). Legacy radial-away/radial-towards and unrestricted free-shell directions are explicit ablations, not thesis-core families. Every row retains strategy provenance, a hard

validity mask, and explicit invalid-reason codes. Branch, beam, and stochastic temperature controls widen data support; they do not replace the RRI objective.

Scale proceeds causally: a small trusted subset first establishes geometry, matching, invalidity, and replay correctness; a scene-level held-out or full ASE GT-mesh subset then supplies thesis evidence; external substrates are eligible only when they preserve target-specific mesh/oracle supervision. Report scenes, snippets, targets, trajectories, anchor poses, candidates, rollout seeds, transitions, split boundaries, invalid gaps, and missing coverage separately. Sample-level splits across snippets from one scene are not sufficient for final claims.

1.2.6 Shared evidence protocol

Invalidity is a shared constraint across all questions. Only physical or explicit admissibility failures (for example collision, out-of-bounds pose, bad frustum, or a contract-defined motion violation) clear action validity; outside-EVL extent, missing semidense/appearance support, and similar evidence gaps remain support fields unless the executable feasibility contract says otherwise. A failed oracle evaluation does not make the action physically invalid: it clears oracle-label validity and Q-training eligibility and is reported as a label/evaluation failure. Keep the mask taxonomy explicit: **action validity** gates selectable candidate rows; **actor evidence/support** records whether observed target or local semidense/EVL evidence exists; **target validity** records whether the target task and identity state are admissible; **oracle-label validity** records whether ground-truth evaluation produced a finite label; and **Q-training eligibility** is the strict intersection required by the learning contract. Report invalid fractions and reason distributions, target-visible fractions, masked and unmasked rank metrics, and selected-action oracle evaluation. Equal budget means equal horizon, candidate count, candidate distribution, and validity constraints; path length, runtime, and oracle calls are separate measurements. Coverage, calibration, stage shift, and storage lineage are evidence qualifiers rather than proxy objectives.

1.2.7 RQ5 — Conditional online discrete Q_H bridge

After offline finite-candidate evidence is stable, does online interaction over the same discrete candidate contract improve endpoint target gain or calibration over the offline Q_H policy? RQ5 reuses the actor-visible state, candidate mixture, hard action mask, target protocol, horizon, and oracle re-evaluation of RQ1–RQ4. It is attempted only when offline headroom and replay support are positive; it is not required to establish the finite-candidate thesis result. Online training must not silently expand the action space, change the target protocol, or treat GT renders as actor evidence.

1.2.8 RQ6 — Lower-priority continuous and simulator escalation

If the finite-candidate and online-discrete evidence is stable, does a continuous or hierarchical target-then-pose policy, potentially in a simulator, provide measurable headroom over the best finite-candidate policy under the same target-specific objective? RQ6 is a time-permitting escalation, not a substitute for the offline Q_H result. Any comparison must preserve target conditioning, matched budgets or cost curves, explicit feasibility handling, and independent oracle endpoint evaluation; no continuous-control result is implied by the current finite-candidate implementation.

Validation TODO: Freeze one claim-level contribution statement for each research question and link it to the exact estimand, evidence artifact, and decision rule that can answer it. Until those links exist, the questions define intended scope rather than validated contributions.

Source: Chapter 5; Chapter 6

Gate: a complete RQ-to-estimand-to-result ledger backed by confirmatory artifacts

1.2.9 Research matrix

Question	Gate	Primary evidence
RQ1	M1, M5	Endpoint target gain, cumulative root gain, diagnostic RRI, and acquisition-cost curves.
RQ2	M4, M5	One-step gate, oracle lookahead, rollout traces, endpoint gain, and recovered headroom.

RQ3	M3, M4	Actor-visible target encoding and leakage-safe crop ablation.
RQ4	M2–M7	Candidate mixtures, branching diversity, validity statistics, scale, and coverage reports.
RQ5	M6, conditional	Online discrete Q_H under the unchanged finite-candidate contract.
RQ6	M6, lower priority	Continuous or simulator escalation against the best discrete policy.

The matrix is a reporting map, not a second source of contracts: implementation semantics remain owned by code, tests, and the linked development reports.

2 Foundations and Related Work

The Introduction located this thesis within active perception: sensing actions change the evidence on which later inference depends [AWB88, Baj88]. Three-dimensional view planning turns that principle into a repeated choice of where to observe next [SRR03]. This chapter develops the concepts needed to make that choice scientifically meaningful before the oracle, data, and model contracts are introduced.

The argument follows the dependencies of the decision problem. It first separates the next-best-view mechanism from the objective used to rank views, then distinguishes target-conditioned reconstruction quality from coverage and uncertainty proxies. It next explains why partial observability and delayed consequences require a finite-horizon information state, derives the geometric properties a candidate scorer should preserve, and finally positions the thesis question against the resulting literature dimensions.

2.1 Active Perception and the Next-Best-View Problem

In active perception, an observation is also an intervention on the future information state. A next-best-view (NBV) method operationalizes this idea by generating possible viewpoints, estimating the benefit of observing from each one, and selecting an action. PB-NBV makes this generate–score–select decomposition explicit for a finite candidate set [Jia+25]. The decomposition identifies three different scientific objects: the proposal mechanism determines which views can be considered, the utility determines how they are ordered, and the selection rule turns that ordering into an action.

How those objects are represented varies across NBV systems. Finite-candidate methods evaluate a bounded set of poses, whereas learned continuous policies predict camera position and orientation directly. GenNBV formulates the latter as a five-degree-of-freedom reinforcement-learning problem, while Hestia factorizes the continuous action into a look-at point followed by a camera position [Che+24, Lu+26]. A finite set makes the evaluated choices explicit and comparable; a

continuous or hierarchical policy can search a broader space but couples view proposal and control more tightly.

These action formulations do not determine what makes a view useful. PB-NBV, for example, accelerates candidate evaluation through projected frontier and occupied evidence, so its proposal and scoring mechanism is tied to a coverage objective [Jia+25]. The same finite candidate table could instead be ranked by uncertainty reduction or by reconstruction error. Consequently, changing the utility changes the scientific task even when candidate generation and selection remain unchanged.

The NBV mechanism therefore answers only **how** a sensing choice is made. To interpret the choice, the utility must state **which reconstruction consequence** is valued and **for which spatial support**. This distinction leads from the general active-perception loop to the objective families compared next [Che+24, Jia+25].

2.2 View Utility and Target-Conditioned Reconstruction

Coverage-based utilities value evidence about previously unobserved surface. SCONE estimates surface-coverage gain from occupancy and visibility fields, MACARONS anticipates coverage gain while reconstructing online, and Hestia rewards newly visible voxel faces [GML22, Gué+23, Lu+26]. These objectives connect a view to geometric completeness, but they weight every counted surface element through the coverage definition; they do not directly ask whether the reconstructed geometry became more accurate.

Information-based utilities instead value a view through its expected reduction of model uncertainty. FisherRF derives an information-gain criterion from the Fisher information of a radiance field [JLD24]. Such a criterion can prefer a view that constrains uncertain model parameters even when its immediate surface coverage is small. The benefit is model-aware exploration; the limitation is that lower parameter uncertainty and lower geometric reconstruction error are related only through assumptions about the model and the evaluated scene.

Direct quality utilities close this proxy gap by evaluating the reconstruction itself. VIN-NBV defines Relative Reconstruction Improvement as the reduction in point-mesh reconstruction error after adding a candidate observation, then learns to rank sampled query cameras by that quantity [Fra+25]. The objective measures the desired geometric consequence directly, but the learned scorer remains myopic: each candidate is valued by the improvement attributed to that next observation rather than by the sequence it may enable.

Target conditioning changes which reconstruction errors contribute to that quality. Object-centric 3DGS planning associates features and confidence with individual Gaussian primitives and gates its information score by the requested object [Jeo+26]. This mechanism demonstrates that a view can be informative for one object without being equally useful for the scene as a whole. It also exposes a separate assumption: supplying an object identifier or region for conditioning is not the same problem as discovering that object and maintaining its identity from observations.

The thesis therefore treats coverage and uncertainty as non-equivalent diagnostics and takes target-specific reconstruction quality as the endpoint of interest [Fra+25, Jeo+26]. This choice fixes what the decision should improve, but not when that improvement should be assessed. Once an action can expose or obstruct later opportunities, view utility must be embedded in a sequential decision problem.

2.3 Sequential Decision-Making under Partial Observability

An egocentric reconstruction system never observes the complete physical scene. Its decision can depend on the sequence of earlier actions and observations, not only on the latest frame. In a partially observable Markov decision process, a belief state is a sufficient statistic of that action-observation history for policy choice [LHP23]. ARIA-NBV does not attempt general belief inference; the relevant consequence is narrower: any finite representation used for view selection is a hypothesis about which parts of the history are sufficient for predicting future target improvement.

The available actions can also change with the information state. A finite NBV method proposes candidates around the current pose and reconstruction, while a hierarchical controller conditions the eventual camera position on an earlier look-at decision [Jia+25, Lu+26]. It is therefore useful to write the admissible choices abstractly as a state-dependent set $\mathcal{A}(s_t)$. This statement excludes unavailable actions from the decision problem; it does not require the Foundations chapter to prescribe how a later implementation stores or learns that exclusion.

Sequentiality matters when the best immediate view is not the best first step of a sequence. For target e , candidate i , and requested horizon h , the finite return and corresponding conditional value are

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

$$Q_{h,e}^*(s_t, i) = \sup_{\pi \in \Pi^{\text{act}}} \mathbb{E}_{\pi} \left[G_{t,e}^{(h)} \mid s_t, a_t = i \right], \quad i \in \mathcal{A}_t, \quad 1 \leq h \leq b_t \leq H_{\max}, \quad Q_{0,e}^*(s, i) = 0$$

where continuation policies choose only from the admissible candidates at each step. Fixed-Horizon TD defines horizon-indexed predictions with the boundary $Q_0 = 0$ and relates horizon h to the shorter horizon $h - 1$ [De +20]. If an episode ends before all h rewards are collected, later rewards are zero. The value thus states the consequence of taking a candidate now and then following a named continuation rule; it is not an intrinsic property of the camera pose alone.

The horizon is distinct from two related bounds. The requested horizon h states how many future acquisition rewards the query represents; the remaining budget b_t limits how many acquisitions are still possible; and $H_{\max}$ marks the largest horizon supported by the chosen model and evidence. Universal Value Function Approximators establish the general possibility of conditioning one approximator on an additional task variable, while Fixed-Horizon TD shares representations across horizon-indexed predictions [De +20, Sch+15]. Neither result licenses extrapolation beyond the horizons actually supported and evaluated.

Learning such values from a fixed data collection introduces an epistemic constraint in addition to the mathematical horizon. Batch-constrained and conser-

vative offline reinforcement-learning methods are motivated by the error that arises when a learned policy assigns high value to actions outside the data distribution [FMP19, Kum+20]. These methods differ in how they control that error, but they share the warning relevant here: a longer horizon does not create evidence for transitions that the data never resolves.

Together, partial observability, state-dependent choices, and finite-horizon return define what a non-myopic scorer must predict [De +20, LHP23]. They do not yet determine how the observation history, target, and candidate geometry should be represented. That question is the next dependency: the representation must retain the distinctions on which future target quality actually depends.

2.4 Egocentric and Geometric Representations

Project Aria records calibrated, time-aligned egocentric sensor streams, and EFM3D lifts posed image features together with semi-dense geometric evidence into a local gravity-aligned voxel representation [Eng+23, Str+24]. This sensing geometry determines how observations from different cameras and times can be compared. It does not, by itself, reveal the complete scene or define which view is useful; an NBV representation must turn the available evidence into relations between the current observer, the target, and candidate viewpoints.

Because the physical scene is only partially observed, representation quality is a question of information preservation. EFM3D supplies strong local evidence but operates over a finite voxel extent, while Hestia’s cumulative voxel-face state shows how viewing direction and observation history can change later decisions [Lu+26, Str+24]. A compact state is therefore not assumed sufficient merely because it is spatial: it must retain target identity, observed support, directional history, and the distinctions needed to compare future candidate consequences.

Coordinate choice controls which distinctions a learner can exploit cheaply. Query-centric models express scene elements in local frames and encode their relations through relative positions, reducing dependence on an arbitrary global origin [Zho+23]. Geometric deep learning generalizes this idea as a choice of

symmetries and locality priors [Bro+21]. For egocentric NBV, translation of the world origin should not change a score, but gravity, metric scale, camera direction, target orientation, and motion can remain meaningful. The appropriate prior is therefore disciplined relative geometry, not automatic invariance to every rigid transformation.

Candidate ordering introduces another symmetry. A candidate table represents a set of physical viewpoints, so permuting its rows should permute the associated scores without changing their values. The required map is therefore permutation equivariant rather than invariant. Deep Sets supplies invariant aggregation for shared context, while Set Transformer supplies equivariant candidate interaction [Lee+19, Zah+17]. This is a behavioral requirement rather than an architecture prescription: independent row scoring, pooled context, or attention can all satisfy it if a row permutation produces the same permutation of the output scores.

The representation family determines how those priors are realized. Point models preserve irregular surface samples and local relative geometry; sparse voxel models provide structured neighborhoods without paying for a dense world grid; equivariant message-passing models impose stronger transformation rules [CGS19, SHW21, Zha+21]. These families trade computational structure against inductive bias. None is an established improvement for this thesis until compared with the same observable inputs, target task, and endpoint utility.

The foundation is thus a set of representational requirements rather than an architecture prescription: causal egocentric evidence, task-relevant spatial relations, candidate-order equivariance, and enough history to distinguish future consequences [Bro+21, Str+24]. These requirements provide the final comparison dimension for the literature synthesis and leave their concrete realization to the Method chapter.

2.5 Related-Work Synthesis and Research Gap

The reviewed approaches differ along four coupled dimensions: what consequence defines utility, whether that consequence is scene-wide or target-specific, which evidence summarizes the acquisition history, and whether the next action is chosen greedily or as part of a sequential policy [Fra+25, GML22, JLD24]. Comparing methods along only one of these dimensions can obscure the scientific difference. In particular, a sequential controller with a coverage reward does not answer the same question as a greedy scorer trained on reconstruction quality.

Family	Utility and target scope	Represented evidence	Decision structure
PB-NBV [Jia+25]	Projected frontier and occupied coverage; reconstruction object	Classified voxel clusters and compact projection proxies	Finite proposed views; greedy score
SCONE / MACARONS [GML22, Gué+23]	Expected surface coverage; scene-wide	Occupancy, visibility, or online reconstruction state	Iterative candidate selection
GenNBV / Hestia [Che+24, Lu+26]	Coverage gain; reconstruction object	Observation history plus geometric or directional state	Continuous or hierarchical sequential policy
FisherRF [JLD24]	Information gain; radiance-field scene model	Model-parameter uncertainty	Candidate views; greedy score
VIN-NBV [Fra+25]	Direct reconstruction improvement; reconstructed object	Candidate-conditioned projections of current reconstruction	Sampled candidates; greedy score
Object-centric 3DGS [Jeo+26]	Object-conditioned information gain; requested object	Per-Gaussian geometry, appearance, and object confidence	Candidate views; greedy score

Table 1: Concept-centred comparison of the literature families that bound the thesis question. Rows are distinguished by utility, target scope, represented evidence, and decision structure; they are not ranked by reported performance across incompatible settings.

The table exposes two tensions that motivate the thesis question. First, coverage and information methods provide mechanisms for exploring missing or uncertain regions, whereas VIN-NBV connects view choice directly to realized reconstruction error [Fra+25, GML22, JLD24]. Second, sequential policies represent observation history and delayed reward, but the compared examples optimize coverage rather than the reconstruction quality of a requested target [Che+24, Lu+26]. Direct quality, target conditioning, and sequentiality are therefore established separately under different assumptions.

This thesis evaluates the bounded conjunction left by those tensions: whether a finite-candidate policy can improve endpoint reconstruction quality for a specified target by valuing more than the next observation, using an egocentric geometric information state [De +20, Fra+25, Str+24]. The claim is a research question, not a conclusion: the literature motivates the combination but does not establish that the available state is sufficient or that non-myopic headroom exists.

The scope remains narrower than a general active-perception solution. The comparison does not claim exhaustive novelty priority, continuous or hierarchical control, actor-visible target discovery, or real-device deployment [Che+24, Eng+23, Lu+26]. Chapter 3 now makes the bounded question operational by separating observable state from the privileged geometry needed to construct target tasks and evaluate reconstruction quality.

3 Oracle and Data Generation

This chapter defines the non-deployable pipeline that turns logged ASE snippets into supervised target-conditioned NBV tasks. It separates actor state from privileged data-generation state, specifies target-task and candidate construction, defines hard invalidity and target-specific RRI, and records the resulting selected counterfactual chains. The learned method in Chapter 4 may consume only the actor-side projection of these artifacts; GT geometry, counterfactual renders, labels, and oracle search remain instruction, supervision, or evaluation assets.

3.1 State and Visibility Boundary

Remove or rewrite before submission: Replace internal protocol and DTO vocabulary such as CF-GT, V1, and storage-shape terminology with scientific concepts in the main argument. Keep exact identifiers only where they provide an auditable implementation anchor.

Source: this section; aria_nbv/aria_nbv/targets/protocol.py; aria_nbv/aria_nbv/data_handling/qh_data/views.py

Gate: a reader can understand the information boundary without repository-local names

Validation TODO: Attach exact local-source locators to the ASE and EFM3D claims, including which logged fields are observations and which geometry products are privileged supervision.

Source: docs/literature/tex-src; docs/references.bib

Gate: claim-level provenance comments resolve every literature claim in this section

The learned component in this thesis is a masked finite-horizon candidate-value function: it predicts Q_H values for a finite candidate table, and the decision rule selects among hard-valid rows. Training evidence is generated by offline, mesh-supervised counterfactual replay. ASE contributes both sensor-like logged streams and privileged ground truth; EFM3D converts the logged subset into a local 3D representation, whereas depth, segmentation, GT boxes, meshes, counterfactual renders, and reconstruction labels remain supervision or evaluation channels [Met25a, Str+24]. Storage in the same adapted snippet does not make these channels equally observable.

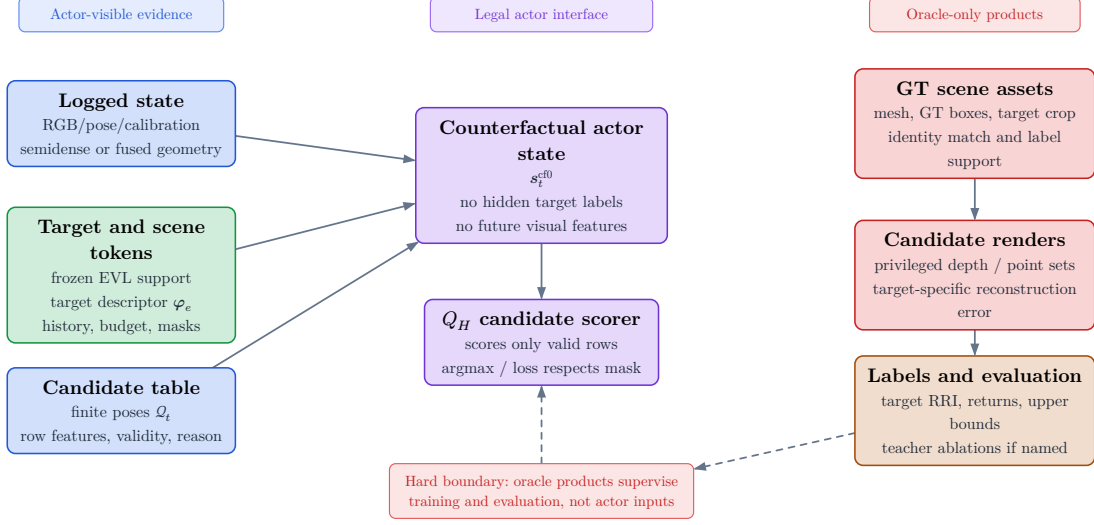


Figure 1: Actor and oracle boundary. Legal Q_H inputs are calibrated logged image evidence, poses, semi-dense geometry with uncertainty and observation support, frozen EVL features or predictions, an explicitly sourced target instruction, selected-view history, remaining budget, candidates, and masks. Reason codes remain audit evidence rather than scorer inputs. GT depth, segmentation, boxes, meshes, target crops, counterfactual renders, labels, and endpoint evaluation remain privileged.

The visibility boundary is protocol-relative and temporal. A dense render for an unselected candidate at the current decision step is oracle evidence. A render from an already selected action may enter a later state only under an explicitly named source protocol: privileged mesh-rendered depth, a declared sensor simulation, or an actor-visible observation. The same array shape can therefore denote different information, and source role must remain explicit.

3.1.1 ASE/EFM Logged Observation State

The target is treated as external task context, so the value is conceptually $Q(s_t, e, a)$ rather than a target-independent state value. The logged historic state contains the recorded trajectory evidence,

$$s_t^{\text{hist}} = (\mathbf{I}^{\text{rgb}}, \mathbf{X}, \mathcal{P}^{\text{semi}}, \mathbf{F}_v, e_t, b_t) \in \mathcal{S}^{\text{hist}}$$

Each component of this tuple has a distinct information role:

Calibrated multi-camera image snippets. EFM3D consumes posed and calibrated RGB and greyscale Aria streams. For every stream and time step, EVL encodes the image with a frozen 2D foundation model, projects the centers of a local 3D voxel grid into the distorted camera model, bilinearly samples the feature maps inside the camera’s valid radius, and aggregates the samples across cameras and time by their mean and standard deviation [Str+24]. In the repository, `EfmCameraView` binds each normalized frame tensor to `CameraTW` calibration, device-clock timestamp, frame identifier, and an optional separately timestamped distance map. The ASE distance map is GT and remains oracle-only [Met25a]. Thus $\mathbf{I}^{\text{rgb}}$ in s_t^{hist} denotes calibrated, time-indexed image evidence; the concrete adapted snippet also carries left and right greyscale SLAM streams rather than a single unposed RGB tensor.

Trajectory, calibration, and gravity. The adapted trajectory is a time-stamped sequence of world-from-rig `PoseTW` transforms with a world-frame gravity vector. EFM3D uses the most recent RGB rig pose, gravity-aligns its rotation, and places a local voxel grid in front of the rig; the per-camera calibration and poses then define the 3D-to-image sampling map [Str+24]. These fields therefore determine the spatial registration, metric scale, gravity direction, and temporal alignment of all lifted evidence. They are also the basis for expressing a candidate relative to the current rig rather than learning values from arbitrary world coordinates.

Semi-dense geometry and observation support. ASE provides MPS-style world-frame semi-dense points, per-point distance and inverse-distance uncertainty, timestamps, and point-to-image observation records [Met25a]. The local `EfmPointsView` preserves padded per-time point slices, valid lengths, metric uncertainty, timestamps, and spatial bounds. EFM3D voxelizes the points into a binary surface mask and uses their observing camera centers to sample the rays before each surface point into a separate free-space mask [Str+24]. Consequently, $\mathcal{P}^{\text{semi}}$ alone denotes sparse surface support; free-space evidence is justified only

when observation lineage and camera poses are retained. A missing point is not evidence that a voxel is free.

Derived EVL field and heads. EVL concatenates the lifted image features with the semi-dense surface and free-space masks, processes the volume with a 3D U-Net, and applies task heads for occupancy and 3D OBB detection. The paper’s detection head predicts voxel centerness, class scores, and gravity-aligned box geometry; its surface head predicts occupancy and is supervised from GT depth samples labelled as free, surface, or occupied [Str+24]. The resulting $\mathbf{F}_v$ is therefore a learned, locally supported statistic of the logged snippet, not another raw sensor modality. The reported EFM3D configuration uses a finite $4\text{ m} \times 4\text{ m} \times 4\text{ m}$ grid anchored at the latest rig pose, so an out-of-extent target or candidate has missing EVL support rather than a measured zero or an invalid pose [Str+24].

The remaining tuple elements are decision context rather than ASE/EFM sensing modalities. e_t identifies the entity whose reconstruction is valued, and b_t conditions the return on the remaining acquisition budget. The current oracle target values are derived from a selected GT box; a deployable target must instead come from EVL predictions or another actor-visible association path. This provenance distinction is consistent with `EfmSnippetView`, which exposes camera/calibration, trajectory, and semi-dense fields as actor observations while keeping `ARIA_OBB_PADDED`, nested `gt_data`, and attached ASE meshes on the oracle/evaluation side.

3.1.2 Counterfactual Actor State

The conceptual counterfactual actor state retains the immutable logged substrate and adds only information causally available after selected actions,

$$s_t^{\text{cf0}} = (\mathbf{F}_v^{\text{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t) \in \mathcal{S}^{\text{cf0}}$$

For the canonical model, this compact tuple is read together with the explicit ordered selected-view history:

$$\mathbf{h}_{t,e,i} = \text{Read}(\Phi_t^{\text{scene}}, \mathbf{h}_e^{\text{tgt}}, q_{t,i}, \mathbf{H}_t, t, H)$$

In $s_t^{\text{cf}0}$, the root EVL field remains fixed, whereas the accumulated point set $\mathcal{P}_t$ may grow with selected observations. The finite candidate table $\mathcal{Q}_t$ contains poses and generator provenance, not observations from those poses. $m_{t,i}$ defines the admissible action set, and the ordered history $\mathbf{H}_t$ retains the selected approach sequence. Invalid-reason codes remain audit evidence and are not scorer inputs.

The implemented profile name `qh_cf0_v1` maps exactly to the weaker **S0-pose** carrier, not to this geometry-updated conceptual state:

$$s_t^{\text{S0-pose}} = (\mathcal{S}_{\text{root}}^{\text{VIN}}, \mathbf{F}_v^{\text{root}}, (\mathbf{T}_{r,e}, \mathbf{l}_e), \mathcal{Q}_t, m_{t,i}, \mathbf{H}_t^{\text{pose}}, b_t)$$

Its actor DTO carries the immutable VIN snippet and compact root EVL evidence, target pose and extent, current candidate poses and action mask, factual selected-pose prefix, and remaining budget. It carries neither the accumulated oracle point set nor invalid-reason codes.

The richer geometry-updated state remains the planned fusion target:

$$s_t^{\text{cf}+} = \left(s_t^{\text{cf}0}, (\mathbf{D}, \mathbf{V}, \mathcal{P}^{\text{cf}}, \mathbf{n})_{1:t} \right)$$

The implemented privileged `qh_cfplus_gt_depth_v1` profile stops at a causal selected-depth carrier:

$$s_t^{\text{CF-GT-carrier}} = \left(s_t^{\text{S0-pose}}, (\mathbf{D}, \mathbf{V}, \mathcal{K}, \mathbf{T}_{\text{root} < \text{-cam}})_{1:t}^{\text{sel}} \right)$$

Selected $\mathbf{D}$ is rendered from $\mathcal{M}^{\text{GT}}$ and paired with its validity mask, candidate-camera calibration, and root-from-camera pose. The implementation does not yet fuse that carrier into $\mathcal{P}^{\text{cf}}$, free-space rays, $\mathbf{n}$, or a refreshed EVL field. Those are planned geometry-state derivations. Even after such fusion, the carrier contains none of the RGB or greyscale images required by EFM3D’s 2D feature encoder and therefore cannot support a fresh EVL field at the selected pose [Str+24].

The transition from t to $t + 1$ may use only the observation produced by the selected row a_t . It may fuse its surface and ray evidence and update support, uncertainty, direction, and recency. It may not attach image features, detections, or EVL descriptors unless the protocol also provides the calibrated camera streams

from which EFM3D derives them. Counterfactual geometry must therefore retain a source role and explicit absence masks for unavailable image-derived channels.

The counterfactual state is thus an **information state** for decision making, not automatically a complete Markov state of the physical scene. It becomes task-sufficient only if its retained root context, causal dynamic evidence, explicit history, target context, action support, and budget preserve every distinction needed to predict future target-specific return. `qh_cf0_v1` is the deliberately weaker `S0-pose` baseline; `qh_cfplus_gt_depth_v1` is a privileged depth carrier, not yet a realization of the geometry-updated state.

3.1.3 Privileged Oracle State

The oracle state adds complete or counterfactual quantities outside the actor input graph:

$$s_t^{\text{oracle}} = (s_t^{\text{cf+}}, \mathcal{M}^{\text{GT}}, \mathcal{M}_e^{\text{GT}}, \mathbf{D}_q, \mathcal{P}_q, \text{RRI}) \in \mathcal{S}^{\text{oracle}}$$

$$\mathcal{M}_{\text{NBV}} = (\mathcal{S}^{\text{hist}}, \mathcal{S}^{\text{cf0}}, \mathcal{S}^{\text{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H)$$

ASE provides per-frame metric GT depth aligned with RGB, per-pixel instance identifiers, class mappings, and the scene trajectory; the EFM3D release adds ASE OBB metadata and validation meshes for object-detection and surface-reconstruction supervision [Met25a, Str+24]. EFM3D uses the depth channel to supervise occupancy at sampled free, surface, and behind-surface points, while visible OBBs supervise centerness, class, and box geometry [Str+24]. These uses establish label provenance, not actor observability.

For ARIA-NBV, $\mathcal{M}^{\text{GT}}$ is the privileged reference surface used for candidate rendering and reconstruction evaluation, while $\mathcal{M}_e^{\text{GT}}$ fixes the selected entity’s evaluation support. All-candidate depth $\mathbf{D}_q$ and backprojected point sets $\mathcal{P}_q$ answer counterfactual queries for unselected rows. RRI then compares target reconstruction error before and after adding that candidate evidence. Depth, points, crops, and RRI are legal for label generation, oracle search, diagnostics, and named upper bounds; none is a contemporaneous input to the student value model.

A GT OBB may define the current oracle identity, pose, extent, crop, and target-bearing instruction. It does not show that the actor detected or associated that object. A deployable protocol must obtain the target hypothesis from EVL predictions or another observation-derived path, retaining ASE identity, segmentation, OBBs, and meshes only for matching and evaluation.

The state hierarchy therefore separates **what was logged**, **what a selected counterfactual history may causally add**, and **what only the oracle can know**. Its layers are related by controlled information projections, not by freely copying arrays between stores. In particular, an oracle tensor may be persisted beside actor tensors for efficient replay without becoming part of the actor state.

3.1.4 Visibility, Feasibility, and Utility

Visibility answers whether a modality contains evidence for a spatial element from a particular view; feasibility answers whether an action is admissible; utility answers how beneficial an admissible action is for the target. These concepts must not be collapsed. A target can be weakly visible from a geometrically valid candidate, a candidate can lie outside EVL support while remaining physically reachable, and an oracle render can fail even though the candidate pose itself is feasible.

Invalidity is a constraint rather than a reward value. Geometry-invalid candidates receive a hard action mask and a persisted candidate reason code before policy selection, stochastic normalization, loss construction, and bootstrap maximization. A geometrically feasible candidate may still have low or negative target gain. Failure of the separate oracle evaluation does not create a candidate reason code: depending on the configured recipe, the affected row or table is skipped, or its oracle-label validity and Q-training eligibility are cleared and the oracle failure is reported separately. Oracle-derived feasibility must be distinguished from a deployable validity estimate when a policy moves from privileged target tasks to an actor-visible target protocol.

3.2 Data Generation and Target-Specific RRI Labels

Validation TODO: Validate the target-cropped error, root normalization, repeatability, failure handling, and endpoint estimand before describing RRI as a reliable study objective. Cite the exact VIN-NBV source lines for inherited definitions and distinguish every ARIA-NBV modification.

Source: docs/literature/tex-src/arXiv-VIN-NBV; metric-validation artifacts

Gate: frozen metric protocol, repeatability table, and claim-level source locators

An oracle target task fixes the entity identity and GT evaluation crop. A deployable target descriptor would instead be predicted from actor-visible observations. The current rollout generator has only the first contract: it projects a selected GT box into a compact instruction for candidate generation. Consequently, the present target descriptor denotes the requested object and its geometry; it does not establish actor-visible target discovery.

The general descriptor notation remains

$$\varphi_e = \text{Enc}_{\text{tgt}}(\hat{\mathbf{B}}_e, \hat{\mathbf{y}}_e, \hat{\pi}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \omega_e^{\text{EVL}}, \ell_e^{\text{src}}, \mathbf{T}_{r_t, e}, \mathbf{T}_{c_t, e})$$

The implemented oracle sampler populates only GT identity, class, confidence, pose, extent, and reference-relative geometry. Projected area, semidense support, EVL support, and proposal-match scores are not measured by this path and must not be interpreted from their schema placeholders.

The finite action interface consists of a candidate table $\mathcal{Q}_t$, hard validity mask $\mathbf{m}_t$, and invalid-reason vector $\boldsymbol{\rho}_t$. The admissible set is $\mathcal{A}_t = \{i : m_{t,i} = 1\}$. Invalid rows remain logged for coverage and failure analysis but lie outside policy argmax, sampling, loss targets, and bootstrap maximization. Low RRI is a valid low-utility outcome; it is never an encoding for infeasibility.

For target e , the oracle computes a target-cropped point-mesh error Δ_t^e . Candidate selection and Q_H supervision use root-normalized target gain; state-relative RRI is retained as a diagnostic. Fixed-budget endpoint gain is the intended policy estimand, but the current replay store records cumulative selected-chain gains rather than an independent post-horizon endpoint reconstruction for every policy. Confirmatory policy comparisons must therefore add matched oracle endpoint re-evaluation or an explicitly persisted endpoint record.

3.2.1 Oracle Label Provenance

The one-step scene-level oracle from the seminar work supplies the rendering and point-mesh substrate [Fra+25, Met25b]. The target-specific pipeline renders valid candidates from the GT mesh, backprojects their depth, fuses candidate points with the privileged current evaluation cloud, crops both points and mesh to the selected target box, and evaluates the resulting point-mesh error. In the current protocol the root evaluation cloud is reconstructed from ASE GT depth, so both the crop and the root metric are oracle-only.

Camera geometry is therefore useful here only as part of a relation: a logged trajectory, a target crop, a finite candidate set, or a rendered oracle query. The calibrated camera boundary is not a contribution by itself. The consolidated candidate-support figure in Figure 3 uses camera glyphs only to expose those relations; native `CameraTW` unprojection and world-from-camera `PoseTW` remain the geometry owners behind the render path.

3.2.2 Target Selection

The implemented sampler does not match actor proposals to GT objects. It enumerates the non-padding GT OBB rows in a snippet, accepts rows with finite positive geometry, and applies seeded uniform sampling without replacement up to the configured per-snippet cap. Thus the stored `matched` status currently means geometry-valid GT task row, not successful proposal-to-identity association. IoU, ambiguity-gap, visibility, and support thresholds are absent from this admission rule.

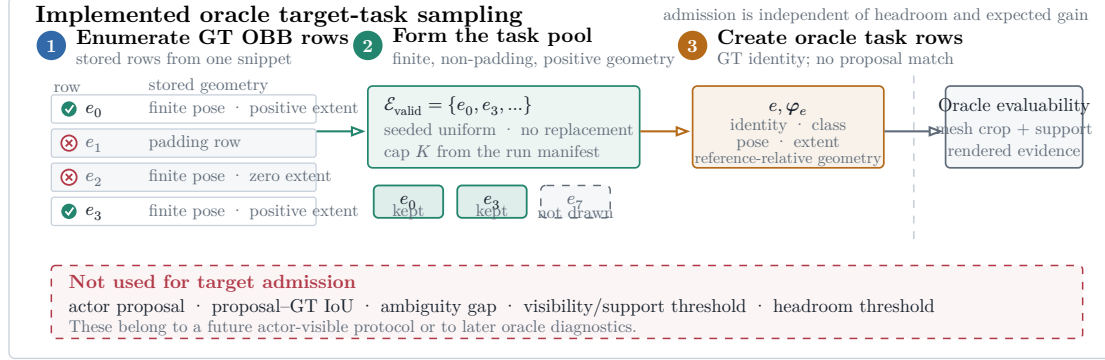


Figure 2: Implemented oracle target-task sampler. Geometry-valid GT OBB rows form the task pool, and seeded uniform sampling without replacement applies the manifest-defined cap. Rollout scoring later decides whether the selected crop is evaluable. No actor proposal, IoU match, visibility gate, or support gate is used.

The sampler bounds rollout cost without pre-filtering on headroom. Candidate scoring may subsequently invalidate the task when its mesh crop, current support, or rendered evidence is unusable; otherwise near-solved and negative-gain targets remain scientifically informative. A later actor-visible protocol must introduce proposal identity and observation-quality diagnostics as a separate selection stage rather than retroactively interpreting the present oracle fields as measurements.

3.2.3 v1 Observed-Target Admission

The implemented v1 audit path keeps this oracle sampler separate from actor-visible target selection. For each observed or predicted target record $\hat{e}$, it compares only same-class GT OBB rows using oriented 3D IoU:

$$\mu_{\text{IoU}}(\hat{e}, e) = \text{IoU}_{3D}(\hat{\mathbf{B}}_{\hat{e}}, \mathbf{B}_e)$$

$$\tau_{\text{IoU}} = 0.20$$

A GT row qualifies only when its IoU is strictly greater than 0.20. The matcher admits the observed target iff exactly one GT row qualifies:

$$n_{\text{qual}}(\hat{e}) = \text{card}(\{e \in \mathcal{E} : \kappa(\hat{y}_{\hat{e}}, y_e) = 1 \wedge \mu_{\text{IoU}}(\hat{e}, e) > \tau_{\text{IoU}}\})$$

$$a_{\text{id}}(\hat{e}) = 1 \text{ iff } n_{\text{qual}}(\hat{e}) = 1$$

Equality at 0.20 is rejected; zero qualifying rows are rejected; and multiple qualifying rows are explicitly marked ambiguous and rejected. A single compatible candidate is admitted only with IoU > 0.20. This implemented **V1** contract has no runner-up-gap or composite- μ claim. The preceding sampler description remains the current **V0** oracle-task path: its geometry-valid GT-row admission is historical **V0** evidence, not actor-visible **V1** matching.

3.2.4 Candidate View Generation

Candidate generation turns one oracle instruction into a finite action table. Every quantitative choice—source population, target cap, shell size, family weights, motion limits, pruning thresholds, rollout recipes, renderer settings, and retention policy—belongs to the resolved run manifest and report bundle. The Methods text defines semantics only; it does not designate one mutable TOML profile as canonical.

At rollout step t , candidate generation constructs a full shell

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = 60$$

with a fixed provenance component $k(i)$ per row. The core mixture contains forward-local, target-bearing, and lateral-bypass motion; the diversity challenger may additionally allocate mass to local refinement and revisit/backtrack components. The resolved manifest, rather than prose, determines which components and counts apply to a run.

For each row, the raw direction is sampled in the reference rig frame. The realistic profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

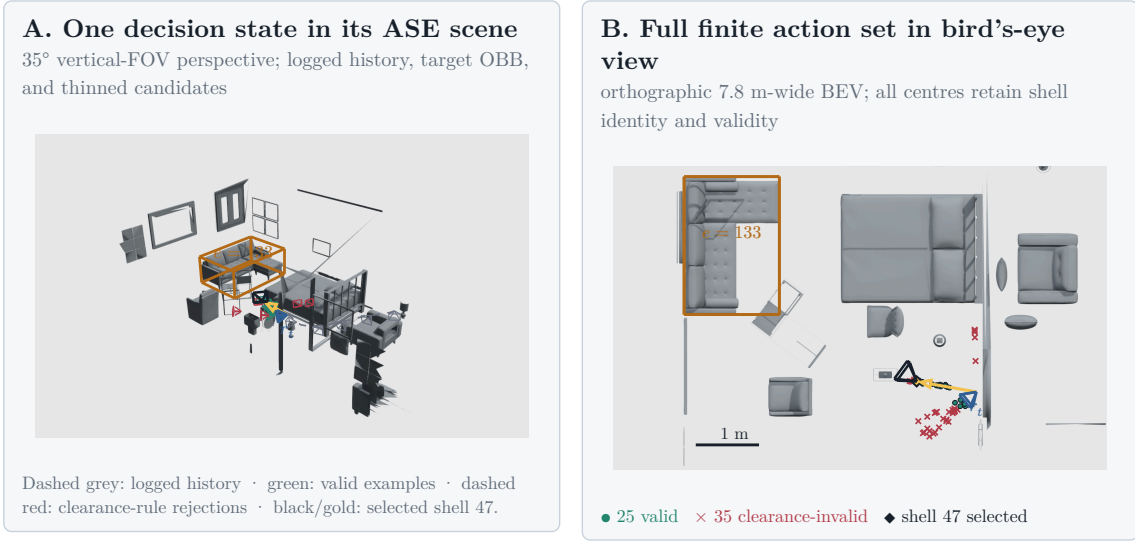
$$\mathbf{u}_i \sim \text{PS}(\mathbf{e}_z, \kappa), \quad p(\mathbf{u}) = c_\kappa (1 + \mathbf{e}_z^T \mathbf{u})^\kappa, \quad \kappa = 8$$

The draw is mapped into configured azimuth and elevation caps without rejection. With $\psi = \text{atan2}(u_x, u_z)$ and $u_y = \sin \theta$, the cap transform is

$$\psi' = \psi \frac{\Delta_\psi}{2\pi}, \quad y' = \sin \theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin \theta_{\max} - \sin \theta_{\min})$$

$$\mathbf{d}_i^0 = \left\| \left(\sqrt{1 - y'^2} \sin \psi', y', \sqrt{1 - y'^2} \cos \psi' \right) \right\|$$

The sampler draws a capped direction in the reference rig frame and reinterprets it as egocentric forward motion, target-bearing motion, lateral bypass, local refinement, or backtracking according to component provenance. Target-looking families orient their optical axis toward the oracle instruction. In this chapter that point is privileged; calling the generator target-conditioned does not make the point actor-visible.



Scene 81286, sample ASE_81286_Atek_000035, rollout row 73, step row 121 24 forward-local + 24 target-bearing
1. 12 lateral-bypass candidates. Dense mesh layers are
z-buffered raster: OBBs, naths, centres, and wire frusta remain vector geometry.

Figure 3: One pinned finite-candidate decision state from ASE scene 81286, sample ASE_81286_Atek_000035, rollout row 73, and step row 121. Panel A uses a 35-degree vertical-FOV perspective view to place logged camera history, root state, target OBB, selected path, and a deterministically thinned set of wire frusta in the real scene. Panel B uses a 7.8-metre-wide orthographic bird's-eye view and retains all 60 candidate centres: 25 rows are admissible, 35 are hard-rejected by the clearance rule, and oracle-greedy selects shell 47. Dense scene geometry is z-buffered; OBBs, paths, centres, and camera glyphs remain vector overlays. Full eye, look-at, up, clipping, and resolution parameters are recorded in the figure JSON. This is an auditable contract example, not a policy-performance result.

The three core position families then reinterpret this capped direction. Let $\mathbf{f} = \mathbf{e}_z$ be the rig-forward unit vector, $\mathbf{b}_e$ the supplied target bearing in the reference

frame, $\mathbf{l}_e = \|\mathbf{e}_y \times \mathbf{b}_e\|$ the horizontal lateral direction, and $\mathbf{e}_y$ the world-up direction expressed in the sampling frame. The family directions are:

$$\begin{aligned}\mathbf{d}_i^{\text{forward}} &= \|\mathbf{f} + \alpha_f(\mathbf{d}_i^0 - (\mathbf{d}_i^0 \cdot \mathbf{f})\mathbf{f})\|, \quad \alpha_f = 0.45 \\ \mathbf{d}_i^{\text{target}} &= \|\mathbf{b}_e + \alpha_t(\mathbf{d}_i^0 - (\mathbf{d}_i^0 \cdot \mathbf{b}_e)\mathbf{b}_e)\|, \quad \alpha_t = 0.4 \\ \mathbf{d}_i^{\text{bypass}} &= \left\| 0.55\mathbf{b}_e + 0.85 \operatorname{sign}(d_{i,x}^0)\mathbf{l}_e + \operatorname{clip}(d_{i,y}^0, -0.35, 0.35)\mathbf{e}_y \right\|\end{aligned}$$

Finally, the sampler draws a radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}(0.25, 1.1), \quad \mathbf{c}_i^w = \mathbf{T}_r^w \left(r_i \mathbf{d}_i^{k(i)} \right)$$

The forward-local family keeps the reference rig orientation. Target-looking families orient the camera toward the supplied target center $\mathbf{p}_e$:

$$\mathbf{z}_i^w = \|\mathbf{p}_e - \mathbf{c}_i^w\|, \quad \mathbf{y}_i^w = \|\mathbf{e}_y - (\mathbf{e}_y^T \mathbf{z}_i^w) \mathbf{z}_i^w\|, \quad \mathbf{x}_i^w = \mathbf{y}_i^w \times \mathbf{z}_i^w$$

These equations describe the sampler, not an optimal proposal distribution. The forward-local family preserves egocentric continuity, the target-bearing-local family moves along the target ray, and the lateral-target-bypass family introduces side-step views; optional challenger families test smaller corrections and reversals. Candidate-profile utility must be judged after pruning. A store in which target-aware families rarely survive cannot support a target-conditioned planning claim.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\mathbf{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max(0, -o_{i,z}) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}$$

The full shell is retained with position, strategy, mixture, sampling probability, rule masks, diagnostics, and invalid-reason bitsets. Panel B of Figure 3 makes the distinction concrete for one stored table: invalid rows remain inspectable but sit outside the admissible set. Invalid candidates cannot enter Q_H selection, stochastic normalization, or loss targets. Conversely, a feasible row with weak target support

or low expected gain remains a valid low-utility example rather than receiving an invalid-reason code. A manifest-defined root-support threshold may reject an entire rollout task when too few actions remain:

$$N_{\text{valid}} \geq \max(12, \text{ceil}(0.25N_q))$$

This threshold is a data-support guard, not a reward threshold. Preflight reporting must retain rejected-root counts and failure reasons by candidate family.

3.2.5 Rollout Branch Sampling and Dataset Impact

Rollout recipes select and retain finite chains from the valid action table. The implemented families are uniform valid sampling, one-step oracle greedy selection, bounded oracle lookahead, and temperature-softmax sampling. Their horizon, branch factor, beam width, temperature, and seed are resolved parameters. These recipes generate replay diversity and bounded references; they do not constitute a learned policy.

Bounded oracle lookahead can select a different first action from one-step greedy because it ranks retained finite-horizon chains rather than immediate gain alone (Figure 4). The persisted artifact contains these selected or beam-retained chains and their full per-step candidate shells; it is not an exhaustive materialization of the counterfactual action tree.

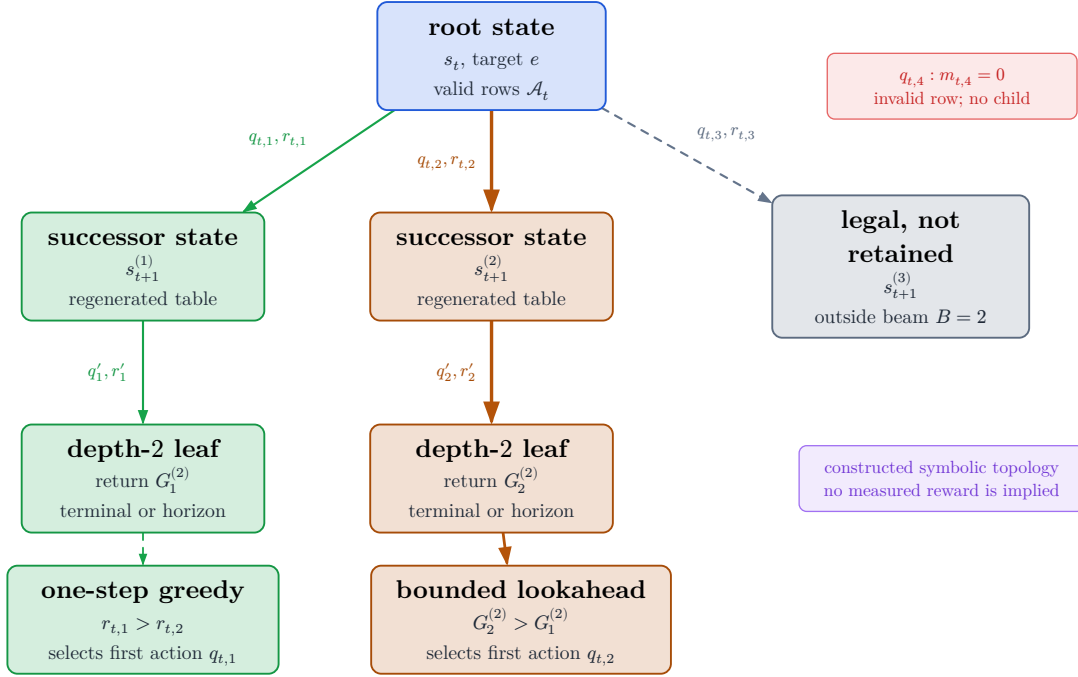


Figure 4: Constructed symbolic topology of the bounded oracle-lookahead reference. Nodes are counterfactual states and edge labels are candidate actions with immediate rewards. Only valid first-action rows produce children; the beam retains a bounded prefix set, and invalid rows receive no branch. The inequalities illustrate how the selected first action can differ from one-step greedy without inventing measured reward values.

For stochastic branches, let s_i be the finite oracle score of valid row i . The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \text{median}(s)}{\text{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1} \exp(\ell_j)}$$

The target source defines the supervised task, the candidate mixture defines learnable action support, validity defines admissibility, and the recipe defines which chains enter replay. Reporting must therefore cover target-task coverage, valid fanout and invalid reasons by family, selected-family diversity, gain distributions, and retention cost before attributing policy behavior to non-myopic planning. The paired train-only pilots are bandwidth and candidate-profile probes; they are non-confirmatory and provide no held-out policy-performance evidence.

3.2.6 Reconstruction-Quality Metric

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated evaluation points to the selected target region. In the thesis protocol, its root points are reconstructed from the observed prefix of ASE GT depth; a candidate contributes renderer-derived depth that is backprojected and fused with the same root cloud. MPS semi-dense points remain actor-visible representation input and a legacy diagnostic source, not the current oracle evaluation cloud.

Writing $P = C_e(\mathcal{P}_t)$ and $M = \mathcal{M}_e^{\text{GT}}$ for the target-cropped point set and mesh, the selected directional accuracy is the mean squared distance from each reconstructed point to its closest GT triangle,

$$D_{P \rightarrow M}(\mathcal{P}, \mathcal{M}^{\text{GT}}) = \frac{1}{\|\mathcal{P}\|} \sum_{\mathbf{p} \in \mathcal{P}} \min_{\mathbf{f} \in \mathcal{F}^{\text{GT}}} d(\mathbf{p}, \mathbf{f})^2$$

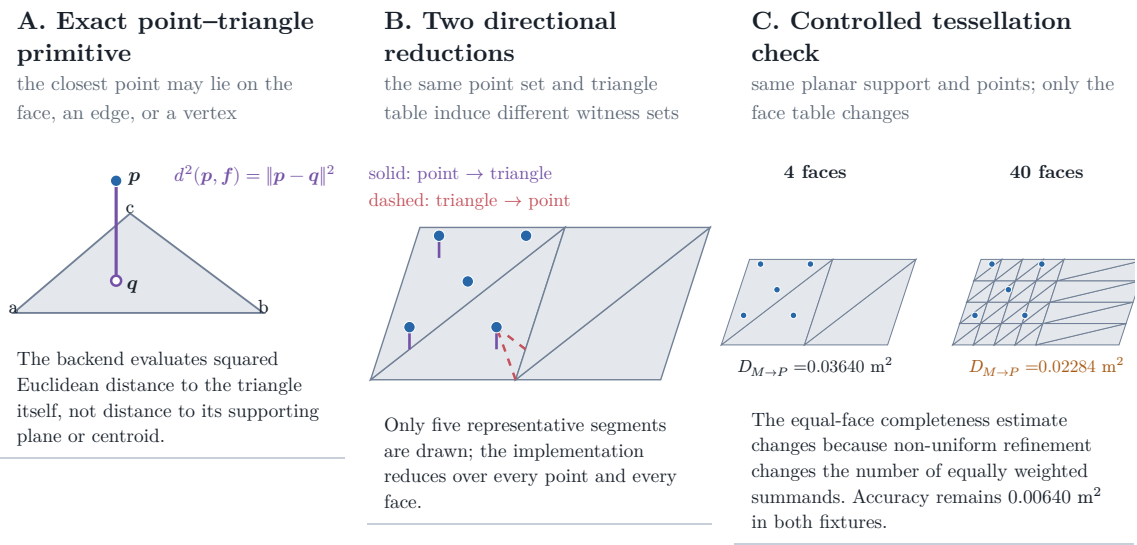
It penalizes reconstructed points that lie away from the target surface and therefore exposes extra, noisy, or misregistered geometry. The reverse term is the mean squared distance from each GT face to its closest reconstructed point,

$$D_{M \rightarrow P}(\mathcal{P}, \mathcal{M}^{\text{GT}}) = \frac{1}{\|\mathcal{F}^{\text{GT}}\|} \sum_{\mathbf{f} \in \mathcal{F}^{\text{GT}}} \min_{\mathbf{p} \in \mathcal{P}} d(\mathbf{p}, \mathbf{f})^2$$

It penalizes target faces without nearby reconstructed evidence and therefore exposes missing support. Their sum is the bidirectional point-mesh error,

$$\Delta_t^e = d(C_e(\mathcal{P}_t), \mathcal{M}_e^{\text{GT}}) = D_{P \rightarrow M, t}^e + D_{M \rightarrow P, t}^e$$

whose directional terms and total have units of square metres. The current and candidate-augmented clouds pass through the same deterministic fusion and point-cap configuration before scoring. This fusion step makes the before/after comparison operationally reproducible; it does not turn the point set into a continuous surface measure.



Controlled validity fixture computed by the repository's PyTorch3D metric and exact Trimesh closest-point witnesses; it is not an ASE performance result.

Figure 5: Point-mesh metric contract and controlled validity fixture. Panel A isolates the exact point-to-triangle primitive. Panel B distinguishes the point-to-face accuracy and face-to-point completeness reductions using computed closest-point witnesses. Panel C holds the planar support and point set fixed while changing only the triangle table; the measured equal-face completeness term changes from 0.03640 to 0.02284 m^2 . These values are generated by the repository's PyTorch3D metric on a synthetic fixture and demonstrate a tessellation-sensitivity mechanism; they are not an ASE performance result.

The completeness reduction gives every mesh face equal weight. It is deliberately neither an area-weighted surface integral nor invariant to retessellation: subdividing part of an unchanged surface changes how much that region contributes. The controlled fixture in Figure 5 demonstrates this property. It bounds the interpretation of the operational estimand but does not invalidate comparisons that share one fixed mesh and scoring contract.

Metric family	Question answered	Dependency and thesis role
Equal-face bidirectional point-mesh error	Are reconstructed points close to GT faces, and does every GT face have nearby point evidence?	Depends on point fusion and mesh tessellation; selected operational RRI error.
Point-sampled Chamfer / accuracy-completeness	Are two sampled surface point sets mutually close?	Depends on sampling density, randomness, norm, and squaring convention; important alternative, not the current metric.
Thresholded precision, recall, and F-score	What fraction of predicted and reference samples fall within tolerance τ , and how do the two fractions balance?	Interpretable at a physical tolerance but threshold-dependent; useful diagnostic rather than the scalar RRI owner.
TSDF, occupancy, coverage, or information gain	How much volumetric surface, free space, or previously unknown space is reconstructed or observed?	Depends on grid, truncation, and visibility definitions; answers a different question from target surface quality.

Table 2: Reconstruction-metric alternatives and their role in this thesis. No alternative is promoted to a co-primary metric.

For comparison, a common squared point-sampled Chamfer convention draws a reference set Q_e from the mesh and averages nearest-neighbour distances in both directions,

$$D_{\text{PS-Chamfer}(\mathcal{P}, \mathcal{Q})} = \frac{1}{\|\mathcal{P}\|} \sum_{\mathbf{p} \in \mathcal{P}} \min_{\mathbf{q} \in \mathcal{Q}} \|\mathbf{p} - \mathbf{q}\|_2^2 + \frac{1}{\|\mathcal{Q}\|} \sum_{\mathbf{q} \in \mathcal{Q}} \min_{\mathbf{p} \in \mathcal{P}} \|\mathbf{q} - \mathbf{p}\|_2^2$$

Its apparent symmetry does not remove sampling-density dependence: changing the number or distribution of samples changes the finite approximation. Some benchmarks also use unsquared distances, so the norm and squaring convention are part of the metric contract rather than an interchangeable naming detail. At tolerance τ , precision counts reconstructed samples with a reference neighbour closer than τ , recall counts reference samples with a reconstructed neighbour closer than τ , and $F_{1,\tau} = 2\text{Prec}_\tau \frac{\text{Rec}_\tau}{\text{Prec}_\tau + \text{Rec}_\tau}$. These scores are physically interpretable but can change with the selected threshold and discard error magnitude on either side

of it. Volumetric coverage and information gain are valuable for exploration and mapping, yet they measure discovered space or uncertainty reduction rather than the target reconstruction-quality change defined here.

The implementation crops mesh faces when any vertex lies inside the oriented target OBB and evaluates the selected point–mesh scorer. Geometry-invalid candidates are removed by the hard action mask and retain persisted candidate reason codes. Empty mesh crops, insufficient current support, or unusable renders instead invalidate the separate oracle evaluation: the recipe either skips the affected row or table, or clears oracle-label validity and Q-training eligibility, while reporting the oracle failure independently rather than assigning a candidate reason code.

This thesis therefore treats the metric as an operational estimand over a frozen evaluation pipeline, not as a representation-independent surface quantity. Every compared policy must share the same target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and metric configuration. A future real-data evaluation that replaces ASE depth with MPS semi-dense evidence must separately quantify its texture-, gradient-, and uncertainty-dependent sampling effects; those concerns are not properties of the current oracle cloud.

3.2.7 Target-Specific RRI

The thesis objective is not the absolute point–mesh error alone, but its normalized reduction after a fixed acquisition budget. It distinguishes state-relative RRI, the additive target-root reward, and endpoint gain: all are dimensionless reductions of the same target-cropped error, but their denominators and scientific roles differ.

The state-relative marginal target RRI remains the VIN-compatible candidate diagnostic:

$$\text{RRI}_{t,i}^e = \frac{\Delta_t^e - \Delta_{t|i}^e}{\max(\Delta_t^e, \varepsilon)}$$

The implementation also reports its selected-chain running sum:

$$C_t^{\text{RRI},e} = \sum_{k=0}^{t-1} \text{RRI}_{k,a_k}^e$$

Because the marginal RRI denominator changes with state, this cumulative diagnostic does not telescope to endpoint improvement. The immediate training reward instead adapts the same error reduction to a target crop and normalizes by the rollout-root target error [Fra+25]:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

With matched sequential states and unit discount, its selected-chain sum is the root-normalized endpoint difference under that same clamped root denominator:

$$J_t^e = \sum_{k=0}^{t-1} r_k^e = \frac{\Delta_0^e - \Delta_t^e}{\max(\Delta_0^e, \varepsilon)}$$

The finite-horizon return is the discounted sum of those target rewards along a selected counterfactual branch. It is a training target for Q_H , not a claim that the deployed system has an online continuous-control policy:

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The log-gain variant is retained only as an internal scale-sensitivity ablation:

$$J_{e,\log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

$J_e^{(H)}$ is the fixed-budget evaluation metric, $G_t^{(H)}$ is the learning return, and $J_{e,\log}^{(H)}$ is a sensitivity diagnostic. The endpoint equation uses $\Delta_0^e + \varepsilon$, whereas the additive target-root reward uses $\max(\Delta_0^e, \varepsilon)$; the two are therefore not canonically interchangeable, even when geometry, horizon, and acquisition budget match. State-relative one-step RRI remains a VIN-compatible diagnostic. The resolved manifest must freeze the discount, clipping, target cap, crop policy, and all evaluation-geometry parameters for each reported experiment.

3.3 Replay Stores and Diagnostic Evidence

Remove or rewrite before submission: Move schema names, directory names, joins, codecs, and DTO-level mechanics that do not change the scientific protocol to the implementation appendix. Keep lineage, missingness, leakage prevention, and reproducibility guarantees in the main text.

Source: this section; `aria_nbv/aria_nbv/rollouts/zarr_store.py`

Gate: every remaining implementation term is necessary to reproduce or interpret the experiment

The pipeline materializes two stores with different ownership. Immutable `vin_offline/` caches logged snippet evidence and expensive one-step oracle products. Standalone `rollouts.zarr/` references those source rows and stores target tasks, retained rollout chains, per-step finite candidate shells, masks, reason codes, selected-action successor evidence, and a derived Q_H view. This separation prevents counterfactual experiments from mutating the source substrate and prevents one-step candidate labels from being mistaken for multi-step replay.

The source store is immutable because it defines the logged actor substrate. Its manifest records source identity, split, schema, materialized blocks, and provenance. A source row may contain frozen EVL/VIN fields, one-step candidate poses and labels, compact rendering payloads, and semidense geometry/history. These fields support scorer training and rollout construction, but they do not themselves define target-conditioned finite-horizon data. Exact directory names, array groups, chunking, and codec choices are versioned reproducibility metadata rather than scientific entities, so they are not reproduced as directory-tree figures in the main text.

The rollout store is normalized around replay identity. One source can produce multiple target rows, each target can produce multiple recipe and retained-chain rows, each chain expands into selected steps, and each step owns a full candidate shell. The store therefore captures selected or beam-retained chain evidence, not the complete counterfactual search tree. Padded `q_h/` arrays are a validated cache over the factual row tables. Invalid candidate rows remain inspectable, but action, training, and bootstrap masks exclude them.

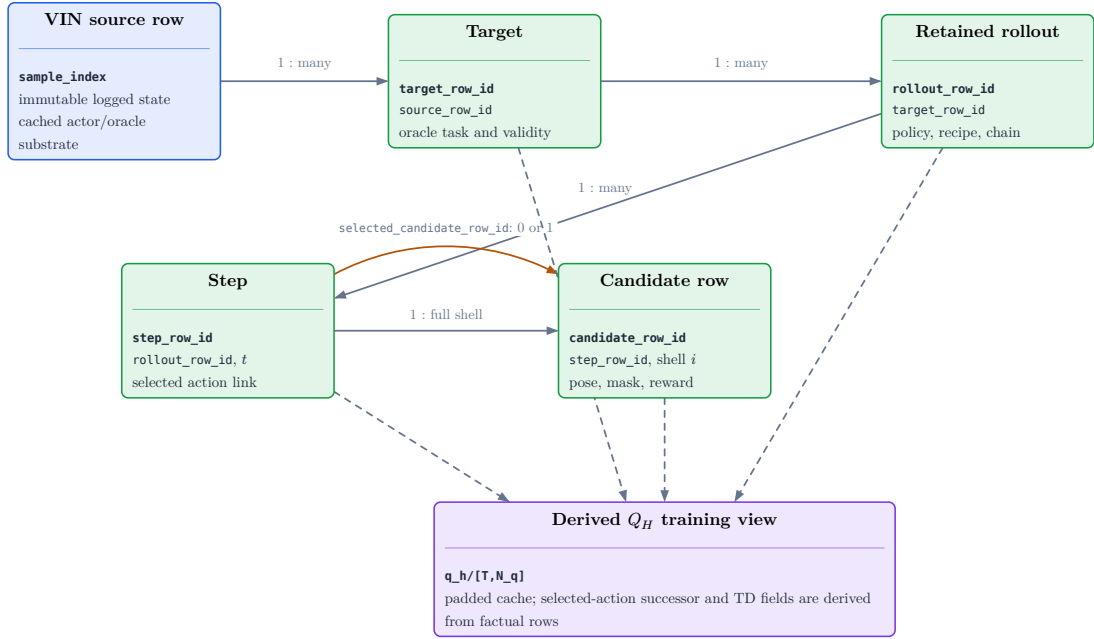


Figure 6: Normalized lineage of the implemented replay evidence. An immutable VIN source row may define several oracle target tasks; each target may produce several retained policy chains; each chain contains ordered steps; and each step owns one full candidate shell. The step row may identify one chosen candidate through `selected_candidate_row_id`; selected-action successor and TD fields are then constructed in the derived $q_h/$ join rather than persisted as a separate transition table.

Evidence family	Interpretation contract
Sources and targets	Manifest-backed task coverage; not proof of actor-visible target discovery.
Candidates and invalidity	Full-shell support with separate hard-action, training, padding, and future deployable-feasibility roles.
Retained chains and steps	Recipe-selected evidence; not a persisted exhaustive search tree.
Selected depth	Chosen-action successor observation with calibration and source role; actor input only under an explicitly admitted later-state protocol.
Q_H view	Derived training cache whose rewards and masks must agree with factual rows; not a scene-memory representation.

Table 3: Interpretation contract for rollout-store audits. Numeric values are rendered from the resolved report bundle in the experiment and reproducibility sections.

Selected-depth persistence stores only the depth raster and calibration for the chosen action at each retained step. It is sufficient to reconstruct the selected-observation prefix without duplicating dense all-candidate renders, but persistence does not decide visibility. A CF-GT reader may use previously selected GT-mesh depths to build a privileged dynamic state; a deployable reader must instead consume a declared sensor-like or observed source. The current unselected candidate renders remain oracle-only in every student protocol. Selected depth is also not an independently scored endpoint artifact.

Likewise, rollout rows summarize final cumulative selected-chain metrics; they do not preserve every rejected branch or a policy-neutral endpoint reconstruction. These limitations must be resolved by matched endpoint re-evaluation before confirmatory policy comparison.

Scientific reporting is generated from the same inspection frames used by the diagnostic application. The main text needs target-task coverage, candidate validity and invalid reasons, family survival and selection, selected-history sanity, gain distributions, source-role counts, and runtime/storage summaries. Exact schema columns, compression, chunking, hashes, and cluster invocation belong in

the reproducibility appendix and must be read from the resolved manifest and report bundle. Development bandwidth pilots are train-only feasibility checks; their counts and throughput may size later jobs but cannot support held-out reconstruction or policy claims.

4 Method

Remove or rewrite before submission: Rewrite this chapter around one frozen, implemented, and evaluated method. Move rejected objectives, alternative carriers, architecture ladders, and unresolved scorer interfaces to development-only notes or a compact limitations/future-work account.

Source: this chapter and its source-owner decision gates

Gate: production scorer, frozen objective, matched controls, and end-to-end acceptance evidence

ARIA-NBV is formulated as target-conditioned selection from a finite candidate table. The implemented substrate generates and evaluates masked multi-step pose rollouts, records selected transitions, exposes a dense training view, and trains an actor-only Q_H scorer through fitted Double-Q optimization. The implemented `A1--S0-pose--root-moments` control independently queries shared scene, target, causal-history, budget, and horizon context for every materialized candidate. It is an executable architecture baseline, not yet a policy result or evidence that the compact state is task-sufficient.

The frozen interface distinguishes remaining budget b_t , scalar requested residual horizon h , and configured support bound $H_{\max}$. The mathematical value family permits $1 \leq h \leq b_t \leq H_{\max}$, but implemented `V1` is deliberately restricted to the trained remaining-budget diagonal $h = b_t$; padding uses zero and every off-diagonal query fails closed. The scorer returns action-mask-independent conditional values and feasibility logits, while Lightning and online inference retain hard-mask ownership. Direct continuous Huber regression is the canonical value objective. A modular CORAL decoder over the same continuous fitted-Q targets is implemented as an ordinal ablation with fixed manifest-bound edges and continuous representatives; exact horizon-two targets, recursive lower-horizon backups, Double-Q selection/evaluation, and behavior-policy returns remain separately named controls or ablations.

The chapter first fixes the actor-visible state and the coverage limitation of local EFM3D evidence. It then separates the implemented compact scorer from richer scene carriers, interaction ablations, and bounded Q_H learning targets. Descriptive status blocks distinguish executable maturity from scientific evidence throughout.

4.1 Actor State and Representation Boundary

4.1.1 Implemented carrier and information boundary

Implementation: partial **Evidence:** pending

The replay carrier, actor/supervision separation, scalar requested-horizon scorer, and fitted-Q adapter are implemented. The current **S0-pose** root-moments state is deliberately compact; task-sufficient dynamic reconstruction memory and frozen scientific validation remain pending.

Literature: [Str+24]

Promotion gate: retain actor/oracle provenance checks and name the admitted state protocol in every run

Development source: `aria_nbv/aria_nbv/data_handling/qh_data/views.py`; `aria_nbv/aria_nbv/rollouts/qh_reader.py`; `aria_nbv/tests/lightning/test_qh_module.py`

The implemented one-step scorer consumes an actor-visible snippet view, a row-aligned candidate table, a reference rig pose, calibrated candidate cameras, and optionally cached EVL output. Oracle RRI labels enter the loss after prediction and are not scorer inputs. The current oracle target descriptor contains semantic identity, pose, positive metric extents, and reference-relative pose in an actor-safe shape, but its values are derived from privileged GT target tasks. A deployable claim therefore still requires observed or predicted target fields with explicit provenance.

The implemented `QhActorTensors` contract carries root semidense evidence, oracle-derived target geometry, root-relative candidate poses, selected-pose history, and remaining budget to the finite-horizon scorer. The current model reduces root evidence to global moments and selected history to a causal pose summary. Candidate regeneration changes pose, history, budget, and the finite action table; richer models must still define how causal selected observations update scene memory without widening this actor/oracle boundary.

Three boundaries are invariant. Invalidity is a hard mask with versioned reason codes, not a small value. GT meshes, target crops, current all-candidate renders, associations, and returns remain oracle or audit fields. Previously selected GT-mesh depth may enter a later state only under an explicitly privileged CF-GT protocol; the corresponding sensor-like or observed variants must use distinct

provenance. Candidate rows remain tied to stable shell identities and documented coordinate frames.

4.1.2 Task-sufficient actor state

Implementation: planned **Evidence:** pending

The canonical state separates immutable root context from a causal dynamic memory. Particular scene carriers are promoted only when they preserve the information boundary and improve target-conditioned decisions.

Literature: [Bro+21, Zah+17]

Promotion gate: selected-observation reader, deterministic fusion, source-dropout tests, and held-out target-RRI ranking

Development source: docs/contents/theory/efm3d_scene_embeddings.qmd; docs/contents/theory/
candidate_view_dependence.qmd

ARIA-NBV does not require one universal reconstruction format. It requires an actor-visible state from which a model can compare a target e and candidate $q_{t,i}$ at decision step t . The frozen scorer interface includes the scalar requested residual horizon h in addition to the factual remaining budget b_t :

$$\mathbf{h}_{t,e,i} = \text{Read}(\Phi_t^{\text{scene}}, \mathbf{h}_e^{\text{tgt}}, q_{t,i}, \mathbf{H}_t, t, H)$$

For architectural and DTO purposes, the resulting scene memory is represented as the state consumed by the value model,

$$\Phi_t^{\text{scene}} = (\mathbf{M}_t^{\text{ray}}, \mathbf{X}_t^{\text{pt}}, \mathbf{F}_t^{\text{DINO@pt}}, \mathbf{E}_0^{\text{EVL-local}}, \mathbf{O}_t^{\text{pred}}, \mathbf{M}^{\text{dir}})$$

The immutable root evidence and causally updated dynamic memory remain separate conceptually: root semidense geometry and supported local EVL features are not silently replaced by selected-observation evidence. The target is external task context in $Q(s_t, e, a)$ and is represented by a separate target state; it is not duplicated inside target-independent scene memory.

The selected-pose history $\mathbf{H}_t$ remains explicit unless a promoted memory is demonstrated to be a sufficient statistic for it. Raw selected depth is an observation consumed by the memory update; it need not remain a direct scorer input once its surface, free-space, support, source, and recency information have been fused.

State protocol	Dynamic evidence	Interpretation
S0-pose	selected poses only	Implemented A1 root-moments control over replay tensors; not a complete reconstruction state.
S1-points	causally fused selected surface points	Minimum geometry-updated counterfactual state; does not distinguish observed free from unknown space.
S2-ray	surface, free, unknown, support, recency	Canonical planned dynamic state for candidate-frustum and target-support queries.
Privileged / sensor-like / actor-visible	source tag on selected observations	Orthogonal information protocol: privileged mesh depth, declared sensor-like simulation, or actor-visible observation.

Table 4: Counterfactual state and source protocols. Scene carrier, information source, interaction architecture, and learning objective are orthogonal experimental choices.

A useful representation must preserve distinctions that can change target-specific return, distinguish missing evidence from predicted free space, and admit a causal update after selection. Candidate queries should read the same memory in target-local, candidate-local, and ray-relative coordinates. A common world translation or yaw convention must not change physical candidate values, while gravity, scale, target extent, camera direction, occlusion, and temporal order remain observable task variables. Exact global $SE(3)$ invariance is therefore neither required nor claimed.

4.1.3 Local EFM3D evidence and the coverage gap

Implementation: partial **Evidence:** pending

EFM3D fields, their voxel pose, and finite extent are persisted. The current finite-horizon scorer consumes lossy global field moments; their sufficiency as the only long-horizon scene representation remains pending.

Literature: [Met25c, Str+24]

Promotion gate: report target and candidate support against the persisted voxel pose and extent

Development source: docs/literature/tex-src/arXiv-EFM3D/method.tex, Sec. Egocentric Voxel Lifting, lines 2–44; docs/contents/literature/efm3d.qmd; docs/contents/theory/efm3d_scene_embeddings.qmd; aria_nbv/aria_nbv/data_handling/vin_store/writer.py

EFM3D encodes logged RGB frames before lifting selected evidence into a finite gravity-aligned voxel field anchored to a snippet pose. The stored voxel transform and extent define the support of lifted features and dense heads. They must not be interpreted as a global volume containing every region observed by the trajectory.

A target or candidate outside this support can remain physically valid; it has missing local-EVL evidence rather than an ordinary zero feature or automatically invalid action.

The coverage limitation motivates a layered interface rather than repeated inference at hypothetical candidate poses. Local EVL features provide high-quality Aria-native evidence where supported. Broader semidense or fused point carriers can preserve observed surfaces beyond that volume. Sparse ray-aware state can distinguish occupied, free, and unknown regions and can be updated from selected geometry. Logged appearance descriptors can be attached to points only after visibility and source lineage are established. None of these carriers creates RGB, DINO, detector, or EVL evidence at an unvisited counterfactual pose.

4.1.4 Ranked representation design space

Implementation: exploratory Evidence: pending

This table ranks scene carriers only. The interaction architecture is specified separately in Section 4.4 so a carrier change does not silently redefine candidate or target semantics.

Literature: [Ave+24, Che+24, Fra+25, Jeo+26]

Promotion gate: promote only after matched support, leakage, runtime, and oracle-policy ablations

Development source: docs/literature/tex-src/arXiv-GenNBV/3-Method.tex, Secs. Formulation and Generalizable State Embedding, lines 10–49; docs/literature/tex-src/arXiv-Hestia/sec/3_method.tex, Sec. Methods, lines 14–58; docs/literature/tex-src/arXiv-scene-script/sections/structured_scene_language.tex; docs/contents/literature/efn3d.qmd

Carrier	Status	Inductive benefit	Cost and promotion gate
Root semidense moments	planned first scorer carrier	Cheap root-level context for contract and optimization tests.	No spatial support or causal update; never interpret as task-sufficient state.
Persisted snippet evidence	available root carrier	Aria-native local features, OBB hypotheses, calibrated cameras, and explicit extent.	Not consumed by a finite-horizon scorer; limited spatial support and requires coverage metadata.
Semidense or fused point memory	first dynamic control	Observed surfaces follow trajectory and selected-view extent.	No free/unknown state; test density and visibility sensitivity.
Sparse ray-aware memory	planned primary extension	Separates surface, free, unknown, support, uncertainty, and causal updates.	Requires deterministic fusion and counterfactual-source masks.
Target-centred EVL re-lifting	adaptation ablation	Reuses logged frame features when the target lies outside the root field.	Domain shift in the 3D neck; compare against simple logged-feature pooling.
DINO-on-point	appearance ablation	Extends logged appearance to observed points beyond the EVL grid.	Needs visibility gating, compression, and missing-descriptor masks.
TSDF/SDF or sparse encoder	geometry/encoder ablation	Compact metric geometry or learned coordinate-bearing tokens.	Must preserve observation weights and unknown-space semantics.
Object-aware 3DGS	renderable-memory ablation	Candidate rendering, soft target membership, and primitive uncertainty.	Per-scene optimization and mask supervision change the state contract.
SceneScript	global-context control	Broad ASE-aligned layout and object hypotheses from semidense input.	Not an ATEK drop-in and does not provide causal free/unknown updates.

Table 5: Ranked scene-carrier design space. Status describes the role in the thesis method, not empirical superiority.

Representation comparisons begin with the smallest planned scorer carrier and then add the smallest carrier that tests a diagnosed information loss. Learned point, sparse, equivariant, or renderable encoders are introduced only when a simpler carrier demonstrably limits target-RRI ranking or finite-horizon recovery. This ordering keeps representation capacity separate from target identity, candidate support, replay validity, and the candidate interaction architecture.

4.2 Descriptor and Encoding Protocol

4.2.1 Persisted replay interface

Implementation: implemented **Evidence:** pending

The factual replay schema, lazy reader, framework-neutral dataset seam, and dense `q_h/` view are implemented and unit-tested. Frozen scientific replay evidence remains pending. The schema is a storage contract; readability from the store does not make a field actor-visible.

Promotion gate: preserve row identity, masks, provenance, and source roles in every tensor reader

Development source: `aria_nbv/aria_nbv/rollouts/zarr_store.py`; `aria_nbv/aria_nbv/rollouts/qh_reader.py`; `aria_nbv/aria_nbv/data_handling/qh_data/dataset.py`; `aria_nbv/tests/rollouts/test_qh_reader.py`

The rollout store preserves source, target, rollout, step, candidate, diagnostic, and lineage tables. Its derived `q_h/` arrays provide a padded state-candidate view without changing factual identities or labels. Target pose and extents in the current oracle task remain privileged ground-truth-derived instructions. The implemented finite-horizon scorer consumes actor-side candidate geometry, selected-pose history, remaining budget, root evidence, and materialization support; target gains, ground-truth associations, mesh diagnostics, crops, current all-candidate renders, and Q labels remain supervision or audit fields. Previously selected depth becomes a later actor input only under a named counterfactual-observation protocol.

Carrier	Persisted content	Learning role
Target	identity, class, pose, extents, reference-relative pose, source and validity provenance	privileged oracle-task instruction; learned actors need observed or predicted equivalents
Candidate	stable row identities, world/root-relative pose, masks, reasons, sampler provenance, support fields	finite action row; privileged diagnostics remain source-gated
Selected chain	selected row, shell index, step order, policy, seed, successor link, terminal state	history and temporal-difference linkage
Selected observation	selected depth, valid mask, calibration, pose and source	later dynamic-state input only under a declared privileged, sensor-like, or actor-visible protocol; never an all-candidate student input
Oracle labels	target RRI, target root gain, errors, optional crops and candidate renders	supervision, evaluation, and audit only

Table 6: Implemented replay carriers and admissible learning roles.

4.2.2 Model-input and DTO contract

Implementation: partial **Evidence:** pending

The implemented DTO seam separates actor inputs, selected-transition supervision, and audit lineage for varying stored chain lengths. The scalar requested-horizon scorer returns a typed conditional-Q/feasibility result, with optional objective-specific CORAL training tensors that never participate in masking. Static scene context, target state, candidate rows, and causal history are implemented; dynamic selected-observation state remains a richer protocol.

Literature: [De +20, Bro+21, Sch+15, Zho+23]

Promotion gate: typed selected-observation state, positive-width actor-visible target path, source masks, leakage tests, and scientific policy evidence

Development source: [aria_nbv/aria_nbv/data_handling/qh_data/views.py](#); [aria_nbv/aria_nbv/lightning/qh_module.py](#); [docs/contents/theory/candidate_view_dependence.qmd](#)

The intended input for target e at step t is

$$\mathcal{I}_{t,e} = \left(\mathbf{h}_e^{\text{tgt}}, \Phi_t^{\text{scene}}, \mathbf{H}_t, \mathbf{b}_t, t, H, \left\{ \mathbf{x}_{t,i}, \mathbf{e}_{a|i}^{\text{rel}} \right\}_{i=1}^{N_q} \right)$$

This equation is an information contract rather than one flat tensor. The corresponding DTO roles are:

DTO role	Candidate content	Visibility
StaticSceneContext	root semidense evidence, supported EVL tokens, root frame and EVL extent	actor input; immutable within one rollout
DynamicSceneState	selected geometry, free/unknown support, recency, source masks and ordered history	actor input; causal update only
TargetState	protocol-specific descriptor, target-local support and field-availability masks	oracle-task instruction or actor-visible target
CandidateTable	row identity, local pose, target relation, actor validity and padding mask	actor input; row-aligned
ValueQuery	scalar requested residual horizon h per state; omission means $h = b_t$	v1 admits only the trained diagonal $h = b_t$; off-diagonal support is planned
CandidateSupervision	one-step root gain, diagnostic target RRI and <code>q_train_mask</code>	supervision only
SelectedTransition	factual action index and row id, reward, discount, terminal and successor identity	training linkage only
AuditLineage	source/store/config hashes, policy, seed and reason vocabulary	CPU audit data; not a learned feature

Table 7: Finite-horizon scorer DTO roles. Model inputs, scalar horizon queries, supervision, transition linkage, and provenance remain distinct even when collated in one training batch.

The maximum supported horizon $H_{\max}$ is a scorer, data, and checkpoint contract. Remaining budget b_t is a factual rollout-state field, whereas requested horizon h selects one member of the scalar family $1 \leq h \leq b_t \leq H_{\max}$. Implemented `bounded_scalar_v1` validates this full syntactic domain: `None` means $h = b_t$, realized off-diagonal calls may request a shorter supported return, and padding alone uses $h = 0$. This admission is not an empirical capability claim. Lightning records the horizons that actually receive targets, and a verified inference bundle rejects any syntactically valid horizon absent from its manifest-bound promoted support. The public output remains $[B, S, N_q]$; multiple horizons use separate scalar calls, while a public vectorized horizon axis remains evidence-gated. The step index t remains lineage by default and becomes a learned feature only in a named non-stationarity ablation.

The executable boundary is `score(actor, requested_horizon=None)`. It hides scene, target, candidate, history, and time encoding behind one deep module and returns conditional Q plus feasibility logits in stored candidate order. The

continuous `conditional_q` field retains one meaning for regression and CORAL: it alone enters Bellman backup and online ranking. CORAL additionally carries cumulative logits and its fixed support as training metadata. Static encodings may be reused privately, but the interface exposes no cache lifecycle, encoder handles, candidate sorting, or public horizon axis.

Padding masks, modality-presence masks, source-role masks, action masks, and training masks remain distinct. Unsupported requested horizons fail closed and are never clamped to the available budget. This does not break recursion from h to $h - 1$: the successor’s factual remaining budget is itself $b_t - 1$, so its diagonal query supplies the required lower-horizon backup. A missing modality must never be encoded as an ordinary zero observation or confused with padding.

The target descriptor separates identity, geometry, observed support, confidence, and source:

$$\varphi_e = \text{Enc}_{\text{tgt}}(\hat{\mathbf{B}}_e, \hat{\mathbf{y}}_e, \hat{\pi}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \omega_e^{\text{EVL}}, \ell_e^{\text{src}}, \mathbf{T}_{r_t, e}, \mathbf{T}_{c_t, e})$$

The learned token combines that descriptor with target-local scene support:

$$\mathbf{h}_e^{\text{tgt}} = \text{MLP}_{\text{tgt}}(\text{concat}(\varphi_e, \mathbf{g}_e^{\text{tgt}}))$$

In the current oracle-task data, target geometry is ground-truth-derived and several generic descriptor fields are unmeasured placeholders. The model contract should therefore use protocol-specific target-state variants, or carry an explicit availability mask per optional field. The same numerical zero cannot simultaneously mean absent support and measured zero support.

Target-independent static scene encodings may be reused across several target tasks. The candidate table, however, is not generally shared: target-bearing, lateral-bypass, and target-looking candidates depend on the selected target. Multi-target evaluation must therefore either carry one candidate table per target or construct a union table with a target–candidate availability mask; only physically identical rows may share candidate encodings.

4.2.3 Relative candidate geometry

Implementation: partial **Evidence:** pending

Root-relative candidate geometry and a minimal candidate-local target relation are implemented. The complete candidate-frustum and dynamic-memory relation descriptor remains planned.

Literature: [Li+21, Zho+19, Zho+23]

Promotion gate: frame-transform, row-shuffle, and held-out descriptor ablations

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/vin/modules/pooling.py

Canonical rigid transforms remain in the store. The reader or tensor adapter derives a candidate pose relative to the current decision reference rather than learning the arbitrary world origin:

$$\mathbf{T}_{r_t,i}^{\text{rel}} = \mathbf{T}_{w,r_t}^{-1} \mathbf{T}_{w,c_{t,i}}, \quad \delta_{r_t,i}^p = \mathbf{R}_{r_t}^\top (\mathbf{c}_{t,i} - \mathbf{c}_{r_t}), \quad \mathbf{R}_{r_t,i} = \mathbf{R}_{r_t}^\top \mathbf{R}_{t,i}$$

$$\mathbf{h}_{t,i}^{\text{pose}}(q_{t,i}; r_t) = \text{concat}\left(\delta_{r_t,i}^p, \text{R6D}(\mathbf{R}_{r_t,i}), \|\delta_{r_t,i}^p\|_2, \text{atan2}(\delta_{r_t,i}^y, \delta_{r_t,i}^x), \Delta h_{t,i}, \mathbf{u}_{t,i}^{\text{up/frustum}}\right)$$

The target relation is encoded separately so candidate self-motion cannot be confused with target conditioning:

$$\mathbf{h}_{t,e|i}^{\text{rel}}(q_{t,i}, e) = \text{concat}\left(\delta_{e|i}^p, \|\delta_{e|i}^p\|_2, \cos \theta_{t,e,i}^{\text{opt}}, \beta_{t,e,i}^{\text{elev}}, \lambda_{t,e,i}^{\text{obb}}\right)$$

Continuous rotation features, metric range, bearing, elevation, height, and frustum variables preserve physical task structure. Query-local relative positional encoding is a later ablation for candidate–target, candidate–history, or candidate–candidate interactions:

$$\delta_{a|i}^p = \mathbf{R}_{t,i}^\top (\mathbf{p}_a - \mathbf{c}_{t,i}), \quad \delta_{a|i}^R = \mathbf{R}_{t,i}^\top \mathbf{R}_a$$

$$\boldsymbol{\eta}_{a|i} = \text{concat}\left(\delta_{a|i}^p, \|\delta_{a|i}^p\|_2, \text{enc}_R(\delta_{a|i}^R)\right), \quad \mathbf{e}_{a|i}^{\text{rel}} = \psi_{\text{rel}}(\mathcal{F}(\boldsymbol{\eta}_{a|i})), \quad a \in \{j, e, k\}$$

This adopts QCNet’s query-centric geometric discipline, not its trajectory decoder or streaming claim [Zho+23]. World-frame copies remain audit facts; the learned interface should expose one authoritative local transform rather than recomputing the same relation through multiple fields.

4.2.4 Candidate rows and directional history

Implementation: partial **Evidence:** pending

The implemented scorer consumes the strictly causal selected-pose prefix through a candidate-relative summary. Ordered token-level temporal encoding and directional memory remain representation ablations for longer-horizon models.

Literature: [e3n25, Lu+26]

Promotion gate: directional-novelty fixture and matched architecture ablation

Development source: [aria_nbv/aria_nbv/data_handling/qh_data/views.py](#); [docs/contents/theory/candidate_view_dependence.qmd](#)

The candidate row is a local query, not a duplicate of the full state. The implemented trunk encodes root-relative and current-relative candidate pose plus global root-scene moments, predicts feasibility before target or horizon conditioning, and then adds a candidate-local target transform for conditional value prediction. Shared target, scene, causal-history, remaining-budget, and requested-horizon context are supplied as state tokens. Hard action validity is not a scorer feature; candidate family or sampler provenance is audit-only by default and may enter the model only as a named ablation.

Viewing history is not reducible to camera distance. For a target-local point or cell, selected camera directions define a signed first moment together with the second-moment memory

$$\boldsymbol{\mu}_t^{\text{dir}}(\mathbf{v}) = \frac{\sum_{k < t} w_k(\mathbf{v}) \mathbf{d}_k(\mathbf{v})}{\sum_{k < t} w_k(\mathbf{v}) + \varepsilon}, \quad \mathbf{M}^{\text{dir}}(\mathbf{v}) = \frac{\sum_{k < t} w_k(\mathbf{v}) \mathbf{d}_k(\mathbf{v}) (\mathbf{d}_k(\mathbf{v}))^\top}{\sum_{k < t} w_k(\mathbf{v}) + \varepsilon}$$

The signed first moment distinguishes opposite approach directions, whereas the second moment records concentration along viewing axes. Their query-relative projections form directional features without committing to a full spherical-harmonic field. Low-order spherical harmonics are promoted only if these moments improve over ordered pose history and still leave systematic directional errors.

Optional appearance, ray, and directional blocks require three independent indicators where applicable: modality presence, batch padding, and evidence source. Removing one optional source must alter only its masked branch, and

counterfactual-only geometry must not receive fabricated RGB, DINO, detector, or EVL descriptors.

4.3 Finite Candidate and Replay Contract

Implementation: implemented **Evidence:** pending

Finite candidate tables, hard masks, lineage, selected transitions, selected-depth persistence, and the derived `q_h/` view are implemented and schema-tested. Frozen scientific store evidence remains pending.

Promotion gate: preserve deterministic shell identity, source roles, and selected-transition validation

Development source: `aria_nbv/aria_nbv/rollouts/replay/types.py`; `aria_nbv/aria_nbv/rollouts/replay/engine.py`;
`aria_nbv/aria_nbv/rollouts/zarr_store.py`; `aria_nbv/aria_nbv/rollouts/qh_reader.py`; `aria_nbv/tests/rollouts/`
`test_qh_reader.py`

At step t , candidate generation returns a finite full-shell table $\mathcal{Q}_t$ with a hard-valid mask $\mathbf{m}_t$ and versioned invalid-reason bitsets. Scores are stored compactly only for hard-valid rows and are bound back to stable shell indices before selection. The admissible action set is

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \{i \in \{1, \dots, N_q\} : m_{t,i}^{\text{act}} = 1\}$$

Candidate orientations factor a component-specific base gaze from a local yaw–pitch perturbation. Production rollout mixtures resolve every component to the final seminar support: symmetric yaw and pitch caps of 60° and 30° , respectively, with zero roll jitter. A zero resolved yaw or pitch cap is rejected as a configuration error. The sampled residuals and their resolved bounds and a per-row bounded-support flag remain attached to the full shell for visual and statistical audit. Legacy zero-cap spherical samplers are explicitly unbounded rather than being misrepresented by a zero-area box; view jitter changes proposal support but does not bypass hard validity rules.

Invalid rows remain available for diagnostics and dense replay, but cannot be selected. A row enters the training mask only when it is actor-selectable and has a finite oracle target. Invalid rows have false masks and undefined labels; scene RRI is never substituted for a missing target-specific label. `valid_action_mask`,

`q_train_mask`, padding, and any deployable feasibility estimate are distinct fields with distinct owners.

4.3.1 Implemented replay transition

Rollout expansion records the full candidate table, selected valid and shell indices, policy scores and probabilities, selection policy, and random seed. The implemented transition is

$$(x_{t+1}, \mathbf{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \text{Step}(x_t, \mathbf{H}_t, b_t, q_{t,a_t}, \xi_t)$$

where x_t is the current reference pose, $\mathbf{H}_t$ the selected-pose history, b_t the remaining budget, and ξ_t the deterministic generation context. The next candidate table is regenerated around the selected pose under the same target task, history constraints, and versioned generator configuration.

This transition is deliberately a replay-control transition, not yet a complete reconstruction-state update. It changes pose, selected-pose history, budget, lineage, and action support. It does not imply that the actor has received a new RGB observation, recomputed EFM3D field, or fused the selected depth into a spatial memory. Any implementation that consumes only these fields must be labelled **S0-pose**.

The persisted factual tables retain source and target identity, lineage hashes, step order, selected rows, masks, reasons, sampler provenance, rewards, selected-depth calibration, and support diagnostics. The derived `q_h/` view right-pads states, stores one-step and target-root-gain labels, and exposes only the factual selected transition needed for a temporal-difference backup. It does not create counterfactual labels for unselected actions or make privileged selected depth actor-visible.

4.3.2 Selected observation and planned scene-state transition

Implementation: planned **Evidence:** pending

A task-sufficient successor state must update only evidence produced by the selected observation. Current GT-mesh selected depth is privileged counterfactual evidence and requires an explicit CF-GT state protocol.

Literature: [Che+24, Lu+26]

Promotion gate: typed selected-observation reader, deterministic fusion, source masks, and no-future-observation tests

Development source: docs/contents/theory/efn3d_scene_embeddings.qmd; aria_nbv/aria_nbv/oracle/evidence.py; aria_nbv/aria_nbv/rollouts/zarr_store.py

A selected observation is a typed tuple containing depth, validity mask, calibration, root-relative camera pose, and an explicit source role. The geometry successor state consumes this selected evidence:

$$s_t^{\text{cf}+} = \left(s_t^{\text{cf}0}, (D, V, \mathcal{P}^{\text{cf}}, n)_{1:t} \right)$$

The source role distinguishes privileged mesh depth, declared sensor-like simulation, and an actor-visible sensor observation. Unselected candidate renders at step t are never elements of the student state.

The existing geometry-level counterfactual uses a set union of retained points,

$$\mathcal{P}_{t+1} = \mathcal{P}_t \cup \mathcal{P}_{q_t}$$

but a planning memory must additionally preserve whether space is observed occupied, observed free, or still unknown. The proposed sparse update is

$$\mathbf{M}_{t+1}^{\text{ray}} = \text{Fuse}(\mathbf{M}_t^{\text{ray}}, \mathbf{P}_{t+1}^{\text{selected-obs}}(a_t), \mathcal{R}_{t+1}^{\text{selected-obs}}(a_t)), \quad a_t \in \mathcal{A}_t$$

Here both evidence terms are indexed by the selected action a_t ; no unselected candidate render enters the successor state. The update may add selected surface evidence, carve observed free space, update support and uncertainty, and refresh target-local directional memory. It must not attach RGB, DINO, detector, or EVL descriptors to counterfactual geometry unless a corresponding actor-visible observation exists. Counterfactual-only cells instead carry explicit source and missing-modality masks.

Raw selected depth is an input to the state builder, not necessarily a permanent model token. The reader may emit either the typed selected-observation prefix or a deterministically derived `DynamicSceneState`, but those two representations must not be mixed silently. This split prevents two common errors: treating pose-only replay as complete visual simulation, and treating privileged selected-depth geometry as a deployable sensor stream.

4.4 Geometric and Mask Acceptance Tests

Remove or rewrite before submission: Retain acceptance properties only for the architecture that is actually implemented and evaluated. Proposed architecture alternatives belong in development notes until a measured failure motivates them.

Source: this section; scorer implementation and tests

Gate: one production scorer passes the stated permutation, masking, frame, and horizon tests

Implementation: partial **Evidence:** pending

The implemented A1 scorer uses independent candidate queries, candidate-relative target/current-pose geometry, scalar requested horizons, and adapter-owned hard masks. Permutation, duplicate, invalid-row isolation, mask independence, and horizon bounds are unit-tested; scientific policy evidence and richer state protocols remain pending.

Literature: [De +20, Bro+21, Lee+19, Zah+17]

Promotion gate: retain permutation, subset, duplicate, padding, mask-independence, frame, source, and horizon tests for every admitted state protocol

Development source: `aria_nbv/aria_nbv/vin/models/target_finite_horizon.py`; `aria_nbv/aria_nbv/lightning/`
`qh_module.py`; `aria_nbv/tests/vin/test_target_finite_horizon.py`; `aria_nbv/tests/lightning/test_qh_module.py`

Candidate order carries no task meaning. For a per-candidate scorer f_θ , jointly permuting row-aligned inputs by Π must permute outputs by the same amount:

$$f_\theta(\Pi \mathbf{X}_t, \Pi \mathbf{m}_t^{\text{cand}}) = \Pi f_\theta(\mathbf{X}_t, \mathbf{m}_t^{\text{cand}})$$

Equivariance alone does not guarantee invalid-row isolation. Holding valid rows and the mask fixed while changing only invalid-row contents must satisfy

$$\text{Mask}(\mathbf{X}_t, \mathbf{m}_t^{\text{cand}}) = \text{Mask}(\mathbf{X}'_t, \mathbf{m}_t^{\text{cand}}) \Rightarrow \text{Mask}(f_\theta(\mathbf{X}_t, \mathbf{m}_t^{\text{cand}}), \mathbf{m}_t^{\text{cand}}) = \text{Mask}(f_\theta(\mathbf{X}'_t, \mathbf{m}_t^{\text{cand}}), \mathbf{m}_t^{\text{cand}})$$

Mask ownership depends on the interaction. In the implemented candidate-to-state scorer, materialized candidates are queries and scene, target, history, budget,

and requested-horizon context are keys and values. `candidate_mask` sanitizes padding before encoding. The scorer does not read `action_mask`; Lightning uses it for Q-loss and bootstrap support, and online inference uses it for the final selectable set. Candidate masks become attention-key masks only in candidate-as-key architectures such as DeepSets context or a masked Set Transformer. Padding, action validity, Q-label support, feasibility-label support, and modality presence remain separate.

Duplicate-row and valid-count tests are required because candidate-set pooling or per-set normalization can otherwise change the absolute value of an unchanged physical candidate. A duplicate row may duplicate an output, but it must not silently change another row’s value unless the tested architecture explicitly models candidate-set context.

Coordinate handling is deliberately weaker than exact SE(3) equivariance. World poses remain available for reproducibility, while model inputs use root-, target-, or candidate-relative geometry. Gravity, scale, height, yaw, camera direction, target orientation, motion limits, and frustum geometry remain physical variables. A global origin convention must not become a shortcut [Bro+21].

Actor/oracle provenance is the final acceptance condition. Target gains, GT associations, mesh distances, current candidate renders, and target crops may supervise or audit a model but may not enter its actor graph. Previously selected GT depth may enter only a CF-GT state branch. Source-dropout tests must show that removing an unavailable optional carrier changes only its masked branch.

The scalar requested-horizon interface adds its own acceptance contract: the mathematical boundary $Q_0 = 0$ is represented only by padded rows in the executable call; an $h = 1$ target contains no bootstrap; $h > b_t$ is rejected rather than clamped; and a target for $h > 1$ may depend only on a successor value requested at $h - 1$ [De +20]. Public vectorization remains gated on real atomic callers, measured inadequacy of private batching, and scalar/vector parity tests. No monotonicity such as $Q_{h+1} \geq Q_h$ is assumed because valid actions may have negative immediate gain.

4.4.1 Orthogonal architecture ladder

Implementation: partial **Evidence:** pending

A1 candidate-to-state cross-attention is implemented with the **S0-pose** root-moments carrier. The identical-feature A0 MLP control, richer scene carriers, and candidate interaction remain orthogonal measurements.

Literature: [Bre+23, Fuc+20, Lee+19, SHW21, Sch+15, Zah+17, Zho+23]

Promotion gate: measure the identical-feature A0 control, then promote a level only after lower interaction controls pass on the same scene carrier and target/source protocol

Development source: `aria_nbv/aria_nbv/data_handling/qh_data/views.py;` `docs/contents/theory/candidate_view_dependence.qmd`

The implemented A1 query contains root- and current-relative candidate pose, an explicit candidate–target transform, and global root-scene moments. Target, scene, causal pose-history summary, remaining budget, and requested horizon are supplied as shared state tokens. The physical trunk and feasibility head precede target/horizon conditioning. Candidate provenance and generator-family identity remain audit-only by default. Ordered history tokens and spatial scene memory replace internal representations behind the same scorer interface when their ablations justify the added capacity [Sch+15].

Level	Interaction model	Scientific role
A0	Independent per-row MLP	Locks target/candidate descriptors, time context, and label learnability for a fixed scene carrier; adapters retain hard masks.
A1	Candidate-to-state cross-attention	Each candidate query independently reads shared target, scene, history, and budget context.
A2	DeepSets candidate context	Tests whether an unordered summary of valid candidates adds information without pairwise attention.
A3	Masked Set Transformer	Tests candidate-candidate interaction while preserving row equivariance and mask isolation.
A4	Query-local relation bias	Tests QCNet-style target, history, and candidate relations without importing its forecasting decoder.
A5	Temporal/recurrent state read	Tests whether ordered long-horizon history remains informative after explicit dynamic scene memory.
A6	Residual value head	Tests finite-horizon recovery over a calibrated continuous one-step root-gain control.
A7+	Exact-equivariant or graph interaction	Escalates only after local-frame controls reveal a symmetry-related failure.

Table 8: Interaction-architecture ladder. A1 is implemented; scene carrier, target/source protocol, time-query contract, and learning target stay fixed for each comparison.

Candidate-to-candidate attention is not required by the core task. It may improve relative policy context or diversity, but unrelated sampled rows must not silently redefine the absolute value of candidate $q_{t,i}$. Exact equivariant layers are similarly scoped to diagnosed support encoders or candidate graphs after local-frame scalar controls. This ordering favors reusable context encodings and keeps scalar requested-horizon conditioning separate from candidate-set interaction.

4.5 Target-Conditioned Finite-Horizon Value Model

Remove or rewrite before submission: Retain scalar requested-horizon conditional Q with direct continuous Huber regression as the core. Keep exact horizon two, CORAL, residual, behavior-return, and alternative estimator variants explicitly labelled as diagnostics or ablations.

Source: this section; `aria_nbv/aria_nbv/lightning/gh_module.py`

Gate: a frozen scorer interface, training objective, checkpoint, and matched policy evaluation

4.5.1 Implemented scorer and training interface

Implementation: partial **Evidence:** pending

The A1-S0-pose-root-moments finite-horizon scorer, typed output, scalar horizon query, feasibility auxiliary, modular regression/CORAL value decoders, fitted-Q learner, and hard-masked online adapter are implemented. The one-step VIN/CORAL scorer remains historical evidence and a myopic control; scientific policy evidence is pending.

Literature: [CMR19, Fra+25, HGS15]

Promotion gate: retain the one-step scorer as a matched control, certify exact Q2, and require held-out oracle-rescored policy evidence

Development source: `aria_nbv/aria_nbv/vin/models/target_finite_horizon.py`; `aria_nbv/aria_nbv/lightning/`
`qh_module.py`; `aria_nbv/aria_nbv/oracle/pipelines/online_qh.py`; `aria_nbv/tests/vin/test_target_finite_horizon.py`

The actor DTO carries root semidense evidence, GT-derived target pose and extent, root-relative candidate geometry, selected-pose history, remaining budget, and candidate materialization support. It deliberately excludes supervision and audit lineage from scorer inputs. The current model makes this an executable S0-pose baseline, but neither a trained checkpoint nor task-sufficient reconstruction state follows from interface tests alone.

The scorer emits two policy-facing tensors before masking:

$$\left(Q_{h,\theta,e,i}^{\text{cond}}, \ell_{t,i}^{\text{feas}}\right) = f_{\theta}(s_t, e, q_{t,i}, h), \quad h = b_t \text{ in implemented V1; } 1 \leq b_t \leq H_{\max}$$

`conditional_q` is finite for every materialized row and is trained only where hard validity and Q-label support admit a target. `feasibility_logits` is produced by a target-, horizon-, and action-mask-independent physical trunk and may receive binary supervision on labelled materialized rows. A configured value decoder may attach training-only tensors, but neither policy masking nor feasibility reads them. Lightning owns hard-mask admission, decoder-specific value loss, optional feasibility BCE, exact $h \leftarrow h - 1$ recursion checks, Double-Q targets, and target synchronization. Online inference compacts conditional values only through the authoritative hard mask.

4.5.2 Scalar requested-horizon scorer

Implementation: implemented **Evidence:** pending

One scalar h per state is implemented. Omitting the query uses factual remaining budget; public horizon vectorization remains evidence-gated.

Literature: [De +20, EGW05, Lee+19, Sch+15, Zah+17, Zho+23]

Promotion gate: exact-Q2 certification, A0/A1 controls, per-horizon support tests, and held-out oracle policy comparison

Development source: docs/typst/thesis/sections/04-method/04-01-scene-representation-requirements.typ; docs/typst/thesis/sections/04-method/04-02-descriptor-and-encoding-plan.typ; aria_nbv/aria_nbv/lightning/qh_module.py

For target e , the return over exactly the requested residual horizon h is

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

and the learned quantity is

$$Q_{h,e}^*(s_t, i) = \sup_{\pi \in \Pi^{\text{act}}} \mathbb{E}_{\pi} \left[G_{t,e}^{(h)} \mid s_t, a_t = i \right], \quad i \in \mathcal{A}_t, \quad 1 \leq h \leq b_t \leq H_{\max}, \quad Q_{0,e}^*(s, i) = 0$$

The notation Q_H names a bounded finite-horizon family. One model query is

$$f_{\theta} \left(s_t^{\text{S0-pose}}, \varphi_e, \{q_{t,i}\}_{i=1}^{N_q}, h \right) \rightarrow \left(\{Q_{h,\theta,e,i}^{\text{cond}}\}_{i=1}^{N_q}, \{\ell_{t,i}^{\text{feas}}\}_{i=1}^{N_q} \right)$$

rather than a value from a separately configured fixed-H model. The maximum $H_{\max}$ is a dataset, model, and checkpoint contract. Requested residual horizon h is a model input and remaining budget b_t determines whether the query is admissible. The step index t stays lineage unless a named non-stationarity ablation uses it.

One candidate row is one geometric query. Candidate pose, candidate-target relation, and root-scene support are encoded once, and a lightweight horizon embedding turns that candidate into a horizon-candidate query:

$$\mathbf{u}_{t,i} = \text{CrossAttn}_{\theta}(\mathbf{x}_{t,i}, \{\mathbf{h}_e^{\text{tgt}}, \mathbf{\Phi}_t^{\text{scene}}, \mathbf{H}_t, \text{Emb}(t), \text{Emb}(H), \mathbf{b}_t\})$$

followed by one shared per-row value head. This conditioning follows the general principle of a shared value approximator queried by an explicit task variable

[Sch+15]. Fixed-H models and separate $Q_1, \dots, Q_H$ heads remain ablations rather than competing public interfaces.

The public interface scores one h per state and returns $[B, S, N_q]$. Multiple horizons use separate calls; private batching may reuse encodings. A public L axis is introduced only after two real atomic callers, measured inadequacy of private batching, and scalar/vector parity tests exist. Every query receives only the causal state available at step t ; the learning target, not future scorer input, determines how many rewards the value represents.

The candidate materialization mask sanitizes padding. Changing only the valid-action mask cannot change raw conditional Q or feasibility logits:

$$\left(Q_{h,\theta,e,i}^{\text{cond}}, \ell_{t,i}^{\text{feas}}\right)(s_t, e, q_{t,i}, h, \mathbf{m}_t) = \left(Q_{h,\theta,e,i}^{\text{cond}}, \ell_{t,i}^{\text{feas}}\right)(s_t, e, q_{t,i}, h, \mathbf{m}'_t)$$

The valid-action mask instead gates the policy argmax, Q supervision, and every bootstrap maximum. Target-independent root encodings may be reused across targets, but target-dependent candidate generators require per-target tables or a union table with an explicit target–candidate availability mask.

The value family is defined relative to a frozen state and source protocol. An **S0-pose** value, a privileged **CF-GT** selected-depth value, and a deployable observed-state value are different functions even when their tensors have similar shapes. Likewise, “optimal” means optimal continuation only within the generated finite candidate support, hard-validity contract, represented state, and transition distribution. A pose-only state cannot be promoted to a task-sufficient reconstruction value merely by requesting a longer horizon.

4.5.3 Horizon-recursive offline learning

Batch fitted Q iteration learns a greedy action-value function from a fixed transition collection through successive supervised regression problems; it does not require online interaction [EGW05]. The frozen lower-horizon recursion is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and, for $h > 1$,

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

where the lower-horizon prediction is detached, frozen, or supplied by a delayed target copy. The essential structural rule for this candidate is $Q_h \leftarrow Q_{h-1}$: no horizon value bootstraps from itself. Fixed-horizon TD motivates this recursion and shows that horizon-indexed values can share parameters and be updated in parallel, although a staged $h = 1$ to H schedule remains the clearest initial control [De +20].

The stored evidence gives a particularly strong base case. Every candidate admitted by `q_train_mask` can supervise continuous one-step root-normalized gain. If a selected first action has a successor table with dense one-step labels, then the exact finite-support $H=2$ target is

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

This exact target is the required recursion check and $H=2$ control. Longer-horizon interpretation remains gated until fitted Q2 matches this factual target on held-out supported rows and oracle lookahead shows positive headroom.

Double Q is an optional estimator for the learned successor maximum. The nonterminal bootstrap first intersects hard action support with factual support at horizon $h - 1$:

$$\mathcal{A}_{t+1}^{(Q,h-1)} = \{i : m_{t+1,i}^{\text{act}} = 1 \wedge m_{t+1,i}^{Q,h-1} = 1\}$$

The online scorer then selects

$$B_t^{(h,e)} = \begin{cases} Q_{h-1,\theta^-,e}(s_{t+1}, \operatorname{argmax}_{i \in \mathcal{A}_{t+1}^{(Q,h-1)}} Q_{h-1,\theta,e}(s_{t+1}, i)) & \text{if } h > 1 \wedge d_t = 0 \wedge \mathcal{A}_{t+1}^{(Q,h-1)} \neq \emptyset \\ 0 & \text{otherwise} \end{cases}$$

and a delayed scorer to evaluate that row. This may reduce overestimation from maximizing noisy learned values [HGS15]. It is neither a requirement for offline learning nor the definition of the frozen scalar requested-horizon interface. It cannot repair unsupported long-horizon actions, an aliased actor state, or missing selected-observation evidence.

Complete stored chains also permit regression to truncated Monte-Carlo returns. Those targets estimate the continuation of the behavior policy that generated each chain, Q^μ , rather than the greedy finite-support value Q^* unless that behavior policy is itself the specified target policy. They are therefore useful controls and diagnostics, but they must not be mixed with optimal Bellman targets without naming the estimand.

The objective-design comparison is therefore:

1. dense continuous Q_1 supervision on every candidate admitted by `q_train_mask`;
2. exact selected-action Q_2 supervision as a base-case certification;
3. direct continuous regression versus fixed-support CORAL decoding over the same fitted-Q targets;
4. the shared scalar-horizon model versus fixed-H or separate-head ablations for $h = 1, \dots, H_{\max}$;
5. Double-Q selector/evaluator backups as a max-bias ablation;
6. behavior-policy Monte-Carlo return regression as a separate estimand;
7. an uncentred one-step-plus-residual decomposition.

Because dense one-step rows vastly outnumber selected transitions at longer horizons, training and evaluation must report support, loss, calibration, ranking, and selected-action regret separately by horizon under either interface. If requested horizons share one learner, their sampling or weighting must be explicit; an aggregate loss must not allow Q_1 to hide longer-horizon failure.

4.5.4 Modular continuous-value decoding

Implementation: implemented **Evidence:** pending

Direct regression and fixed-support CORAL are executable alternatives over the same candidate-state features and the same continuous fitted-Q targets. Regression remains canonical; CORAL is an ordinal ablation, not a new action-value estimand.

Literature: [CMR19, Dab+17]

Promotion gate: fit-data-only support selection, held-out per-horizon ranking and calibration, support-saturation reports, and matched policy evaluation

Development source: `aria_nbv/aria_nbv/vin/modules/qh_value_decoders.py`; `aria_nbv/aria_nbv/lightning/qh_module.py`; `aria_nbv/tests/vin/test_qh_value_decoders.py`

The representation trunk produces one feature vector per materialized candidate row. A terminal decoder maps that vector to the scalar `conditional_q` consumed by the unchanged Double-Q backup and hard-masked selector. The regression decoder applies a per-row MLP and linear output, trained with Huber loss in continuous root-gain return units. The CORAL decoder applies a per-row MLP and $K - 1$ cumulative thresholds. For fitted-Q target y_n , fixed edges e_k^Q create the ordinal class c_n^Q standard cumulative binary cross-entropy supervises the thresholds [CMR19]:

$$c_n^Q = \sum_{k=0}^{K-2} \mathbb{1}[y_n > e_k^Q], \quad l_{n,k} = \mathbb{1}[c_n^Q > k]$$

$$\mathcal{L}_Q^{\text{CORAL}} = - \sum_n \sum_{k=0}^{K-2} (l_{n,k} \log p_{n,k} + (1 - l_{n,k}) \log(1 - p_{n,k})), \quad p_{n,k} = \sigma(o_{n,k})$$

$$\pi_{n,k}^{\text{raw}} = p_{n,k-1} - p_{n,k}, \quad p_{n,-1} = 1, \quad p_{n,K-1} = 0$$

$$\tilde{\pi}_{n,k} = \frac{\max(\pi_{n,k}^{\text{raw}}, 0)}{\sum_{j=0}^{K-1} \max(\pi_{n,j}^{\text{raw}}, 0) + \varepsilon}$$

$$Q_n^{\text{cond}} = \sum_{k=0}^{K-1} \tilde{\pi}_{n,k} u_k^Q$$

CORAL provides order, not metric distance. The continuous interpretation used by backup and ranking is an additional experiment contract: repaired class mass is

averaged through fixed, strictly increasing representatives u_k^Q . Edges, representatives, threshold initialization, and decoder kind are therefore scorer configuration and inference-bundle identity. Changing them defines another model even if K is unchanged. Invalid candidates are not mapped to the lowest class; hard-invalid and unsupported selected rows are excluded before either Huber or CORAL loss.

The outer CORAL classes are open-ended while decoded values saturate at the outer representatives. Every CORAL run consequently reports the fraction of fitted-Q targets below and above the representative support, the outer-class fraction, and pre-repair cumulative-probability order violations. Support edges and representatives must be selected using fit data only and then frozen before validation or test evaluation. A one-epoch GPU smoke proves the executable transaction and bundle reconstruction, not scientific superiority or comparability of Huber and cumulative-BCE loss magnitudes.

If a third decoder is promoted, quantile regression is the most coherent next study: it estimates a return distribution whose expectation can preserve the scalar ranking seam, whereas CORAL only discretizes one scalar target [Dab+17]. Such a head is justified only after stochastic returns or actor-state aliasing are measured, and it requires a separately frozen distributional Bellman projection, quantile support, risk-neutral or risk-sensitive decoding rule, and calibration evaluation. It must not be introduced as a stylistic replacement for direct Q.

4.5.5 One-step base and finite-horizon residual

Implementation: exploratory Evidence: pending

The residual decomposition is a testable hypothesis. Direct continuous finite-horizon value prediction remains the required control under whichever time-query interface is selected.

Literature: [De +20, CMR19]

Promotion gate: target-specific root-gain calibration and direct-versus-residual ablation across horizons

Development source: docs/literature/tex-src/arXiv-VIN-NBV/sec/3_methods.tex, ordinal RRI paragraph, line 125; docs/contents/theory/candidate_view_dependence.qmd

The hypothesis separates calibrated immediate utility from downstream effects:

$$b_{\psi,i} = f_{\psi}^{1\text{-step}}(s_t^{\text{cf0}}, \varphi_e, q_{t,i}), \quad \delta_{\theta,t,e,i}^h = g_{\theta}(\mathcal{J}_{t,e}, q_{t,i}, h), \quad Q_{h,\theta,e,i} = b_{\psi,i} + \delta_{\theta,t,e,i}^h$$

The base $b_{\psi,i}$ must predict continuous one-step target root gain

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

in the same additive units as the finite-horizon return. Historical state-relative RRI or an ordinal CORAL score may remain auxiliary ranking and calibration outputs, but they are not interchangeable with this additive base unless an explicit calibrated conversion is learned and validated. The residual captures candidate regeneration, selected-observation state changes, occlusion, support, and the requested residual horizon.

It is not exactly mean-centred within each candidate table because duplicate or unrelated rows would then change absolute value targets. Magnitude regularization may be tested without redefining the value field:

$$Q_{h,\theta,e,i} = \hat{r}_{\psi}^e(s_t^{\text{cf0}}, \varphi_e, q_{t,i}) + \delta_{\theta,t,e,i}^h(\mathcal{J}_{t,e}, q_{t,i}), \quad \mathcal{L}_{\delta} = \lambda_{\delta} \frac{1}{|\mathcal{A}_t|} \sum_{j \in \mathcal{A}_t} (\delta_{\theta,t,e,j}^h)^2$$

The historical one-step CORAL score remains a motivated myopic ranking and calibration interface, but it must not be confused with the implemented finite-horizon CORAL decoder above. The historical head discretizes one-step RRI; the new decoder discretizes the continuous fitted-Q target and decodes back to return units. Fixed-H and requested-horizon models, the exact H=2 control, Double-Q ablation, Monte-Carlo control, and residual learner form separate comparisons rather than one implicit architecture.

Research TODO: Compare staged and joint shared-parameter variable-horizon fitted Q against separate per-horizon heads; include dense Q1, exact Q2, Double Q, behavior-return regression, and the uncentred residual after positive oracle-lookahead headroom is established.

Source: variable-horizon learning-target contract

Gate: A0/A1 learning, per-horizon support report, and headroom report

Open decision: Freeze the maximum horizon, horizon-sampling or loss-weighting rule, lower-horizon target-update schedule, and target-network rule in the resolved training manifest. For the residual ablation, also record whether the one-step base is frozen, slow-fine-tuned, or jointly trained.

Source: variable-horizon fitted-value hypothesis

Gate: model-selection protocol freeze

The scientific endpoint is not training loss. For every claimed horizon, selected trajectories are regenerated under the documented candidate and state protocol and re-scored by the same target-specific oracle used for the baselines. The model succeeds only if it recovers a prespecified fraction of positive oracle-lookahead headroom on held-out scenes without violating horizon, mask, provenance, source, or support constraints.

5 Experimental Design

The experiments separate feasibility evidence from policy evidence. Feasibility requires a frozen scene-level population, reproducible target and candidate support, valid replay linkage, and measured oracle throughput. Policy inference additionally requires a held-out scene split, matched acquisition budgets, independent endpoint oracle evaluation, and an implemented actor-visible decision rule. The current train-only bandwidth pilots address only the first category: incomplete runs and resource failures characterize the generation system, but they cannot support comparisons between policies.

5.1 Study Population and Evidence Gates

Validation TODO: Preregister the eligible population, exclusions, primary estimands, aggregation unit, number of independent runs, uncertainty interval, minimum meaningful effect, and multiplicity policy before inspecting confirmatory results.

Source: experiment manifest and analysis specification

Gate: immutable analysis plan plus matched held-out policy table

The source population comprises ASE/ATEK snippet windows from scenes for which the configured GT mesh and object-box table resolve. A frozen manifest assigns entire scenes to train, validation, or test before model selection; no scene may cross these boundaries through another snippet. Each reported run records the manifest hash and the exact counts of scenes, snippets, admitted target tasks, rollout chains, transitions, and retained candidate rows. A capped train-only pilot is therefore a throughput and support probe, not a sample from which held-out policy performance can be estimated.

The present data generator defines oracle target tasks by seeded sampling from geometry-valid GT OBB rows. Its task-coverage report therefore describes the available GT pool, sampled tasks, classes, scenes, and later oracle-evaluation failures. It does not measure proposal matching, IoU ambiguity, projected visibility, or actor-observation support. Those quantities belong to a future observed-target selector required for deployable-input claims. Until that selector exists, GT target geometry may define labels and bounded oracle references but cannot be

presented as actor-visible input. The privileged-supervision boundary in Figure 1 applies equally to render-derived evidence: dense GT candidate depth may produce labels, returns, or explicitly named privileged ablations, but it is not a legal actor input. A separate teacher policy, distillation path, or current-belief renderer remains a hypothesis until implemented and evaluated, so none is promoted to a main-text process figure.

The first policy gate is an actor-visible myopic scorer over the same finite candidate table intended for Q_H . The scorer must expose one value per candidate, respect the hard action mask, and be assessed by candidate ranking, calibration, and oracle-rescored selected actions. The existing scene-level VIN scorer is historical substrate; it is not a target-conditioned control until the observed target descriptor is wired into the model and evaluated on a frozen held-out split.

The second policy gate estimates whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only if the preregistered analysis classifies this headroom as meaningful is Q_H evaluated for recovery from offline rollout traces:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is measured by matched endpoint oracle evaluation, not predicted values or training loss. If lookahead has no meaningful headroom, the result is scoped to the frozen split, target protocol, candidate generator, horizon, branch factor, and validity regime. If headroom exists but the learned model does not recover it, target observability, action support, replay coverage, reward construction, and model capacity remain separate candidate explanations.

Claim	Primary evidence	Decision rule
Population	scene-split manifest and coverage bundle	Inference is restricted to the frozen held-out scene population.
Task protocol	GT pool, sampled tasks, classes, and oracle failures	Current evidence is oracle-task coverage, not observed-target matching.
Myopic control	ranking, calibration, and oracle-rescored selections	Actor-visible target conditioning must be implemented before comparison.
Planning headroom	Δ_{look} and recovered fraction η_Q	Q_H is evaluated only after a meaningful-headroom gate.

Table 9: Objective-to-evidence matrix.

5.2 Replay Eligibility and Finite-Horizon Learning Gate

The factual rollout tables determine which records may supervise each learning problem. The action-validity mask defines actions selectable under the admitted oracle geometry contract. The stricter Q-training eligibility additionally requires a valid target/ground-truth label state and finite target-root-gain and diagnostic target-RRI labels. Padding, actor validity, one-step training eligibility, transition eligibility, modality presence, source role, and horizon availability remain separate masks. Masked rows remain available for support and failure analysis but cannot enter action selection, supervised loss, or bootstrap maximization.

All eligible candidate rows can support dense one-step supervision. Exact H=2 supervision is narrower: the factual first action must have a stored reward and either an explicit terminal outcome, whose continuation is exactly zero, or a valid nonterminal successor step whose candidate table exposes at least one finite one-step root-gain label. General recursive supervision is narrower again: it requires a factual selected action, reward, terminal flag, discount, a defined training horizon, and—when nonterminal—a reproducible successor state, hard mask, and lower-horizon factual support. The derived $\mathbf{q_h/}$ arrays align these fields on a padded state-candidate view; they do not create labels for unobserved transitions, make selected GT depth actor-visible, or turn sparse long-horizon action support into dense support.

Learning surface	Purpose	Minimum factual evidence
dense $h = 1$	immediate candidate value	actor-selectable row with finite one-step root-gain label
exact $h = 2$	base-case finite-support value	selected reward plus either terminal flag or a nonterminal successor table with at least one finite one-step root-gain label
recursive $h > 1$	variable-horizon fitted value	selected transition, terminal and discount; nonterminal rows also require successor actor state, hard mask, and lower-horizon target support
behavior return	policy-conditioned Monte-Carlo control	complete retained reward prefix and behavior-policy identity

Table 10: Required replay evidence by learning target. Dense immediate labels, exact $H=2$ targets, recursive optimal-continuation targets, and behavior-policy returns are distinct supervision surfaces.

Gate	Available evidence	Required before inference
Oracle data	target labels, masks, lineage, replay validation, and selected-depth persistence	held-out coverage, source-role audit, and matched endpoint evaluation
Myopic control	scene-level VIN substrate	actor-visible target-conditioned Q_1 scorer and frozen checkpoint
Finite-horizon scorer	A1-S0-pose/root-moments scalar-horizon scorer and fitted-Q seam	exact-Q2 certification, compatible checkpoint, frozen state protocol, and oracle-rescored policy
Requested horizons	fail-closed scalar query and exact $h \leftarrow h - 1$ recursion tests	supported targets through $H_{\max}$, positive headroom, and per-horizon validation
Dynamic Q_H	selected-observation persistence and planned state update	typed dynamic-state reader, deterministic fusion, source masks, and held-out policy evaluation
Policy claim	train-only feasibility pilots	completed held-out paired comparison under equal budget

Table 11: Learning-readiness gates. A runnable scorer establishes executable readiness, not a scientific policy result.

Implementation: partial **Evidence:** pending

The scalar requested-horizon A1 scorer, hard-masked selected-transition Double-Q learner, feasibility auxiliary, and exact-Q2 diagnostic are implemented. Task-sufficient dynamic state and policy evidence remain unavailable.

Literature: [De +20, EGW05, FMP19, HGS15, Kum+20]

Promotion gate: exact Q2 certification, supported H>2 targets, compatible checkpoint, frozen state protocol, and held-out oracle re-evaluation

Development source: `aria_nbv/aria_nbv/data_handling/qh_data/batching.py`; `aria_nbv/aria_nbv/lightning/qh_datamodule.py`; `aria_nbv/aria_nbv/lightning/qh_module.py`; `aria_nbv/aria_nbv/rollouts/qh_reader.py`

The finite-candidate value model decodes actions only over valid candidate rows:

$$\mathbf{u}_{t,i} = \text{Enc}_\theta(\mathcal{J}_{t,e}, q_{t,i})$$

$$a_t^\theta = \underset{i: m_{t,i}^{\text{act}}=1}{\text{argmax}} Q_{h,\theta,e}(s_t, i)$$

The masked argmax is already the discrete decision rule. A separate actor network and online data collection are not required to train or execute this finite-candidate policy. Batch fitted Q iteration explicitly learns a greedy Q function from a fixed collection of transitions by repeatedly solving supervised regression problems [EGW05].

5.2.1 Scalar requested-horizon targets

The scorer represents $Q_{\theta(s_t,e,i,h)}$ for each residual horizon h admitted by $1 \leq h \leq b_t \leq H_{\max}$. The boundary target is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and the recursive target is

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

The lower-horizon prediction is treated as a fixed regression target by stop-gradient, a frozen stage checkpoint, or a delayed target copy. The defining recursion is $Q_h \leftarrow Q_{h-1}$ rather than $Q_h \leftarrow Q_h$, and the executable learner rejects a successor whose factual horizon is not exactly $h-1$. Fixed-horizon TD was introduced

precisely for predictions over a bounded number of future rewards and avoids same-horizon self-bootstrapping; its horizon functions may use shared parameters and parallel updates [De +20].

The clearest initial schedule is staged backward induction with one shared horizon-conditioned network:

1. fit Q_1 from dense one-step labels for every candidate admitted by `q_train_mask`;
2. freeze or snapshot the lower-horizon target path;
3. fit Q_2 from selected transitions and validate it against the exact dense-successor target;
4. continue through Q_H , always requesting $h - 1$ from the successor target path;
5. optionally fine-tune all horizons jointly after the staged model passes per-horizon regression and ranking gates.

This schedule preserves one inference interface and shared encoders while making target lineage explicit. Fixed-H networks and separate per-horizon heads are matched ablations, not alternate runtime contracts.

For remaining horizon two, the store supplies an exact target whenever the successor table has dense one-step labels:

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

This target uses no learned successor value or target network. Agreement between fitted Q_2 and this exact control is a required base-case test before interpreting longer-horizon results under either scorer interface.

5.2.2 Double-Q and behavior-return controls

Double Q changes how a noisy learned successor maximum is estimated; it does not define the scalar horizon interface. The online path selects

$$B_t^{(h,e)} = \begin{cases} Q_{h-1,\theta^-,e} \left(s_{t+1}, \operatorname{argmax}_{i \in \mathcal{A}_{t+1}^{(Q,h-1)}} Q_{h-1,\theta,e}(s_{t+1}, i) \right) & \text{if } h > 1 \wedge d_t = 0 \wedge \mathcal{A}_{t+1}^{(Q,h-1)} \neq \emptyset \\ 0 & \text{otherwise} \end{cases}$$

and the delayed path evaluates $Q_{|\theta)}(s_{t+1}, e, j^*, h - 1)$. This selector/evaluator split can reduce overestimation caused by maximizing noisy action values [HGS15]. It remains an ablation against the simpler frozen lower-horizon maximum. It is relevant in an offline setting only because a learned maximum is present, not because online learning is planned.

A retained chain also yields the truncated Monte-Carlo target

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

for its behavior policy μ . Regression to this fixed target is a useful policy-conditioned control, but it estimates Q^μ , not the greedy finite-support value Q^* , unless μ is explicitly the target continuation policy. Behavior returns from random-valid, greedy, softmax, and oracle-lookahead chains must therefore remain identified rather than pooled as if they represented one optimal value function.

Double Q addresses one maximization bias. It does not create missing successor transitions, make unsupported actions reliable, or repair state aliasing. CQL and BCQ motivate explicit offline-support diagnostics because a greedy learned policy can select actions whose multi-step consequences are weakly represented in the behavior data [FMP19, Kum+20]. Conservative regularization is introduced only if those diagnostics reveal systematic unsupported-action overestimation.

5.2.3 Horizon-balanced replay and evaluation

Dense one-step rows vastly outnumber selected-action targets at longer horizons. An unweighted row mean can therefore optimize the myopic task while obscuring longer-horizon failure. The implemented learning contract freezes this aggregation as `uniform-admitted-selected-row-huber-mean-v1`, binds it with the maximum horizon, replay candidate/rollout configuration identities, behavior-policy vocabulary, reward/return semantics, and discount, and rejects cross-stage mismatches. Checkpoint admission binds the actor-state identity separately. Alternative sampling or weighting schemes must receive new versioned learning identities. Candidate alternatives are:

- a horizon-stratified sampler;
- per-horizon loss weights w_h ;
- or a fixed number of admitted targets per horizon and scene.

Every run reports, separately for each h :

- admitted states, selected actions, behavior policies, candidate families, scenes, and targets;
- value loss and signed target error;
- candidate ranking and top-action regret where oracle comparison is available;
- bootstrap, terminal, and no-valid-successor fractions;
- online/target disagreement for Double-Q runs;
- marginal selected target RRI and the selected-chain cumulative target RRI diagnostic;
- cumulative target root gain, kept distinct from cumulative RRI;
- endpoint performance of the masked policy requested at the remaining budget.

A single scalar validation loss is insufficient for model selection unless its horizon aggregation is frozen in advance. Cross-stage corpus admission enforces the same maximum horizon, reward and return semantics, discount, state/source protocol, candidate/reason vocabulary, replay-support identity, and horizon-weighting rule.

The optimality claim remains bounded under either interface. A learned finite-horizon value can approximate the best continuation only within the sampled finite candidate generator, hard-validity regime, represented actor state, and offline transition support. The same checkpoint cannot silently mix a pose-only state with privileged selected-depth, sensor-like, or actor-visible dynamic states. Longer horizons increase—not decrease—the need for selected-observation geometry and a sufficiently Markov scene state.

The current scorer-independent learner establishes infrastructure readiness only. Confirmatory interpretation requires the source-owner scorer decision, a canonical oracle-task corpus, dense-Q1 and exact-Q2 controls, supported targets for every claimed horizon, cross-stage learning-contract equality, checkpoint compatibility, a frozen actor-state protocol, horizon-specific support diagnostics, and

matched held-out endpoint oracle re-evaluation. A dynamic-state claim additionally requires selected-observation fusion and no-future-observation leakage tests.

5.3 Policy Comparison and Statistical Protocol

Implementation: partial **Evidence:** pending

The aggregation and artifact-driven reporting seam exists, but confirmatory paired policy evidence does not.

Promotion gate: frozen held-out manifest, completed paired endpoints, and validated report bundle

Development source: `aria_nbv/aria_nbv/rollouts/reporting.py`; `docs/typst/thesis/experiment_data.typ`

The experimental unit is the scene. Every comparison is paired by scene, target task, root state, candidate-generation distribution, hard validity regime, and acquisition horizon. Planner depth may differ because it defines the decision rule, but every policy receives the same acquisition budget. Repeated snippets, targets, and seeds are aggregated within scene before the primary comparison so that densely sampled scenes do not dominate the estimand.

The primary estimand is the paired per-scene difference in fixed-budget endpoint target-quality gain, using $J_e^{(H)}$. Its $\Delta_0^e + \varepsilon$ denominator is distinct from the $\max(\Delta_0^e, \varepsilon)$ denominator of cumulative target-root gain, while state-relative RRI has a state-varying denominator; both remain secondary diagnostics rather than alternate names for the endpoint. The centralized analysis artifact freezes the scene aggregation function, uncertainty procedure, interval level, comparison set, and minimum meaningful effect before final aggregation. Invalid-action rate, runtime, path length, and task/candidate coverage likewise diagnose mechanism and feasibility but do not replace the endpoint estimand. The current replay store alone cannot estimate this endpoint comparison because it does not persist an independent post-budget reconstruction for every policy; confirmatory analysis therefore requires matched endpoint oracle re-evaluation or an equivalent frozen endpoint record.

Support and failure strata are fixed before policy inspection. The report distinguishes source rows without an admitted target task, admitted tasks whose

oracle evaluation fails, roots without a valid action, trajectories that terminate before the shared budget, and completed paired trajectories. A policy that cannot continue remains in the comparison with no further gain; it is not silently dropped. These strata delimit the analysis population and expose whether a policy difference is instead a support or systems failure.

Policy	Decision information	Acquisition budget	Scientific role
π_{rand}	actor-visible	H	valid-action lower reference
$\pi_{\text{learned-1}}$	actor-visible	H	learned myopic control
$\pi_{\text{oracle-1}}$	oracle immediate reward	H	one-step oracle-greedy comparator
$\pi_{\text{oracle-look}}$	bounded oracle lookahead	H	conditional finite-support upper reference
π_Q	actor-visible	H	planned learned finite-horizon recovery

Table 12: Matched policy comparison. All rows acquire the same number of views; oracle access and planner depth define the decision rule rather than the acquisition budget.

The first inferential comparison is bounded oracle lookahead against one-step oracle greedy. Its evaluation contract freezes one target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and point-mesh metric configuration for every policy. Repeated evaluation must reproduce the same scores within the declared numerical tolerance; it verifies deterministic execution under this contract, not invariance to a different mesh, sampling process, or metric. The analysis artifact also freezes the meaningful-headroom rule before learned-policy inspection. Recovered-headroom ratios are reported only when their denominator passes that gate. The learned comparison then contrasts Q_H with the actor-visible myopic control under the same endpoint evaluation and acquisition budget; because neither learned target-conditioned control is presently complete, this comparison remains prospective.

Validation TODO: Populate the evidence chain in order: evaluation-contract identity and deterministic repeatability, candidate and target support, oracle-lookahead headroom, myopic-control calibration, finite-horizon recovery, then representation and architecture ablations. Missing upstream evidence blocks downstream claims rather than becoming a zero result.

Source: thesis objective-to-evidence contract

Gate: artifact-backed Results bundle

Validation TODO: Run row-shuffle, mask-isolation, duplicate-row, valid-count, frame-transform, target-source-dropout, and horizon-boundary tests for every model admitted to the policy comparison.

Source: geometric acceptance contract

Gate: architecture validity report

Open decision: Freeze the scene aggregation, interval procedure, interval level, comparison family, and meaningful-headroom threshold in the resolved analysis manifest rather than in thesis prose.

Source: artifact-driven reporting plan

Gate: confirmatory analysis freeze

5.4 Outcome Logic

Meaningful, stable lookahead headroom establishes only that the frozen finite-candidate setup contains exploitable non-myopic structure. No meaningful headroom is a setup-specific negative result, not evidence that target-aware planning is universally myopic. A non-reproducible or mismatched evaluation contract, or insufficient paired support, blocks downstream policy claims. Conclusions remain conditional on the frozen metric contract and are not claims of representation independence. Likewise, train-only bandwidth pilots and failed generation attempts, including memory-exhaustion failures, inform compute configuration and throughput planning only; incomplete artifacts cannot establish headroom or policy superiority.

6 Results

Validation TODO: Replace fixture- and readiness-oriented output with confirmatory results that answer each research question using matched endpoint evaluation, denominators, exclusions, aggregation units, independent-run uncertainty, and immutable artifact provenance.

Source: confirmatory report bundle and analysis manifest

Gate: submission evidence bundle passes all report assertions and every stated result resolves to raw and derived artifacts

The thesis report bundle is loaded through the strict schema checked in `experiment_data.typ`. Its declared evidence class is *pilot*. Schema validity establishes only that provenance, missingness, and table contracts are readable; it does not turn a development fixture or an incomplete pilot into scientific evidence.

Result	Required evidence	Status
Population and targets	population facts with denominators in every validated store	not available
Candidate support	valid/total candidates and zero-action failures in every validated store	not available
Paired policy effect	effect, interval, scene count, and headroom gate in every validated store	not available
Resource feasibility	wall time, peak GPU memory, and storage in every validated store	not available

Table 13: Evidence availability in the loaded report bundle. “Not available” means that no thesis result is inferred from fixture values, train-only attempts, or an incomplete store.

6.1 Candidate and Store Feasibility

The available artifacts show that the finite-candidate rollout path reaches mesh rendering, target-specific oracle scoring, and selected-action replay on training sources. They therefore support an implementation-readiness claim only. A CUDA out-of-memory failure in an unbatched candidate render and later memory-bounded attempts identify rendering as a scale gate; neither establishes rollout throughput, storage cost, candidate-family support, or policy quality for the intended study population.

6.2 Target-Task Coverage

The implemented pipeline samples geometry-valid GT target tasks and records target and validity fields in the rollout schema. The loaded bundle does not contain a validated population and exclusion table for the study. Scene coverage, eligible-target coverage, invalid-reason frequencies, and their denominators are consequently unavailable as results.

6.3 Policy Comparison

The primary estimand is not estimable from the current evidence because no validated held-out bundle contains the matched policy outcomes and aggregation inputs. Oracle repeatability, oracle-lookahead headroom, learned one-step performance, finite-horizon recovery, uncertainty intervals, and sensitivity analyses are therefore not reported.

6.4 Runtime and Storage Gate

The runtime and storage claim requires completed validated stores whose resolved manifests, failure records, and resource summaries refer to the same run. Until that evidence exists for both rollout profiles, the large-scale data-generation gate remains pending. The current attempts justify memory-bounded rendering and retained failure provenance, but no extrapolation to cluster throughput or dataset volume.

7 Discussion

Remove or rewrite before submission: Rewrite the discussion after confirmatory results exist. The present implementation-readiness narrative and architecture bridge registry are development material; the final chapter must interpret measured answers, alternative explanations, scope, limitations, and failure modes.

Source: Chapter 6 and the frozen analysis plan

Gate: every interpretation points to a reported result or is explicitly bounded as a limitation

The current evidence supports a narrow implementation conclusion. ARIA-NBV has an executable finite-candidate path that separates actor-visible policy inputs from oracle-only target geometry, applies invalidity as a hard decision constraint, evaluates target-specific reconstruction change offline, and exposes the resulting store through one typed reporting seam. This establishes that the proposed experiment can be represented and audited; it does not establish that one candidate family, rollout policy, representation, or learned model performs better than another.

The rollout attempts also expose a systems limitation. Counterfactual scoring repeatedly renders a large mesh for a candidate set, so renderer memory and latency constrain feasible branch factor and rollout volume before statistical efficiency becomes relevant. An out-of-memory observation is evidence for batching and resource measurement, not for or against the candidate distribution or planning objective. Completed validated stores are required before throughput, storage, or failure rates are generalized.

The scientific limitations are more restrictive. Current attempts use training sources, no held-out paired policy table supports the endpoint estimand, metric repeatability and meaningful oracle-lookahead headroom remain unestablished, and the target-conditioned finite-horizon comparison has not yet been demonstrated. The present evidence therefore cannot distinguish negligible non-myopic structure from inadequate candidate support, target observability, replay coverage, representation support, or model failure.

7.1 Evidence-conditioned architecture bridges

Implementation: exploratory Evidence: pending

Alternative representations and policies remain part of the thesis design registry, but they are promoted only as responses to diagnosed bottlenecks.

Promotion gate: a measured limitation that the proposed bridge directly tests

Development source: Table 5; Table 8

If the local EFM3D field is the limiting variable, broad semidense memory, sparse occupied/free/unknown state, target-centred re-lifting, or logged appearance features provide progressively stronger representation tests. If compact descriptors lose relevant spatial structure, point, sparse, or equivariant encoders become justified. If independent candidate queries leave errors correlated with the valid candidate set, DeepSets or masked candidate interaction becomes relevant. If selected-view history fails specifically by approach direction, signed first- plus second-moment memory, followed only if needed by spherical harmonics, becomes the targeted ablation. A renderable 3DGS state is appropriate only when candidate rendering or soft instance membership is itself the measured missing capability.

Continuous and hierarchical policies change the action contract rather than merely increasing model capacity. A target-then-pose policy could first select or mask a target token and then condition pose queries on target, horizon, and step; multiple targets can share scene encodings and be evaluated in parallel. This remains post-core work because continuous pose generation requires separate feasibility, safety, support, and simulator-realism evidence. The finite-candidate model is retained as the calibrated control even if a continuous policy is later introduced.

Sequence or recurrent models are similarly conditional. They become scientifically motivated when errors persist after explicit history, time, horizon, and scene-memory conditioning, not simply because the dataset contains trajectories. Looping or recurrent refinement must preserve causal masks and candidate-row equivariance and must be compared against the same fixed-budget endpoint oracle evaluation.

Research TODO: After the primary evidence chain is complete, promote only bridges tied to a measured failure: EVL support, target observability, directional history, valid-set interaction, long-horizon credit, or action-support mismatch.

Source: method design-space registry

Gate: artifact-backed failure analysis

The next evidential step remains procedural: complete validated stores under resolved manifests, freeze the eligible population and exclusions, establish oracle stability and headroom, train the matched controls, and only then interpret architectural differences. The registry above preserves concrete follow-up hypotheses without presenting them as validated conclusions.

8 Conclusion

Validation TODO: Rewrite the conclusion as direct, evidence-calibrated answers to RQ1–RQ4. The current conditional outcome tree is useful development guidance but is not a final scientific conclusion.

Source: Chapter 6; Chapter 7

Gate: confirmatory results and discussion support one concise answer per research question

This thesis defines a leakage-auditable experiment for target-conditioned finite-candidate next-best-view planning. Its present contribution is the separation of actor-visible state from oracle supervision, the target-specific reconstruction objective, hard validity and replay contracts, and an artifact-driven reporting seam that keeps provenance and missingness attached to later results.

The available evidence does not answer whether bounded oracle lookahead improves fixed-budget target reconstruction or whether a learned finite-horizon policy recovers such headroom. The current training-source rollout attempts establish pipeline reachability and reveal a renderer resource gate; the development report fixture establishes the data contract only. Neither supports a held-out policy claim, a population-level effect, or a scale estimate.

The final scientific conclusion is therefore conditional on evidence that is not yet available. A stable oracle metric and positive paired lookahead effect would define measurable headroom for the evaluated finite support. Negligible headroom would be a setup-specific negative result. Unstable oracle evaluation would block planning claims, and stable headroom without learned recovery would remain a learned-control failure with several unresolved mechanisms. These outcomes delimit the thesis without extending it to continuous control, online reinforcement learning, or real-device deployment.

Development roadmap

This page records schedule, status, evidence pointers, and promotion gates. It is omitted from submission output. The active thesis owns claims; Python, tests, configuration, and immutable artifacts own behavior and measurements.

Implementation: **partial** **Evidence:** **pending**

Planning status only; this view promotes no policy-improvement claim.

Promotion gate: M1 correctness and M5 headroom evidence

Development source: docs/typst/thesis/sections/; docs/typst/thesis/development/ml-contract-report.typ; aria_nbv/; aria_nbv/tests/; active configuration

Outcome

The development outcome is a reproducible, evidence-backed thesis package. Canonical narrative and equation labels live in docs/typst/thesis/sections/; this page tracks readiness and gates only.

Milestones

- **M0 — scope and foundation (2026-04-29–2026-05-10): status: complete** gate: thesis scope, source policy, and citation owners aligned. Evidence: docs/typst/thesis/main.typ, docs/typst/thesis/sections/01-research-questions.typ, docs/references.bib.
- **M1 — data, cache, and oracle correctness (2026-05-11–2026-05-31): status: blocked** gate: fresh store, scene split, frame/depth, candidate alignment, Rerun, and throughput evidence. Evidence: docs/typst/thesis/development/ml-contract-report.typ#ssec:m1-status, aria_nbv/tests/data_handling/, aria_nbv/tests/pose_generation/, aria_nbv/tests/rollouts/, .configs/offline_only.toml.
- **M2 — VIN baseline and scale gate (2026-06-01–2026-06-21): status: blocked by M1** gate: reproducible baseline, diagnostics, and scale receipt. Evidence: aria_nbv/tests/vin/, aria_nbv/tests/lightning/, docs/typst/thesis/sections/05-experimental-design/05-01-objectives-and-hypotheses.typ, docs/typst/

thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ.

- **M3 — target-RRI and generation readiness (2026-06-22–2026-07-12): status: blocked by M1** gate: trusted target subset, actor-visible protocol, and deterministic generation checks. Evidence: docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ, aria_nbv/tests/oracle/test_target_selection.py, aria_nbv/tests/pose_generation/, aria_nbv/tests/rollouts/.
- **M4 — target-conditioned one-step scorer (2026-07-13–2026-08-09): status: pending M3 evidence** gate: held-out ranking, calibration, oracle selections, and grouped failures. Evidence: docs/typst/thesis/sections/04-method/04-02-descriptor-and-encoding-plan.typ, aria_nbv/tests/oracle/test_scoring.py, aria_nbv/tests/lightning/test_candidate_scorer_contract.py.
- **M5 — lookahead headroom and Q_H (2026-08-10–2026-08-30): status: pending prerequisite gates** gate: equal-budget rollout comparisons, measured headroom, and supported Q_H evidence. Evidence: docs/typst/thesis/sections/04-method/04-05-finite-candidate-value-model.typ, docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ, aria_nbv/tests/data_handling/test_qh.py, aria_nbv/tests/rollouts/.
- **M6 — scale and bridge decisions (2026-08-31–2026-09-13): status: pending M5** gate: scale decision and explicit bridge scope. Evidence: docs/typst/thesis/sections/07-discussion.typ, aria_nbv/tests/oracle/test_online_vin.py, aria_nbv/tests/rollouts/test_reporting.py.
- **M7 — final experiments and writing (2026-09-14–2026-09-27): status: pending M5/M6** gate: immutable run/config links, coverage statement, final tables, and failure cases. Evidence: docs/typst/thesis/sections/06-results.typ, docs/typst/thesis/sections/07-discussion.typ, aria_nbv/tests/rollouts/test_reporting.py.

- **M8 — release freeze (2026-09-28–2026-09-30): status: pending** M7 gate: reproducible configs, docs, demo path, smoke checks, and completed HM/FK07 pre-submission checks. Evidence: Makefile, docs/README.md, CI workflows, final evidence bundle, and authenticated PRIMUSS registration records.

Ablations

Comparison axes remain in canonical owners; this page records pointers and gates only.

- **Target input:** docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ; gate: V0/V1 protocol tests and actor-visible boundary.
- **Candidate mixture:** docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ, aria_nbv/aria_nbv/pose_generation/; gate: validity and provenance diagnostics.
- **Objective/planner:** docs/typst/thesis/sections/05-experimental-design/05-01-objectives-and-hypotheses.typ, docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ; gate: equal budget and oracle receipt.
- **Invalidity/learning:** docs/typst/thesis/sections/04-method/04-04-architecture-contract.typ, docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ; gate: masks, reason codes, replay tests.
- **State/scale:** docs/typst/thesis/sections/04-method/04-01-scene-representation-requirements.typ, docs/typst/thesis/sections/03-oracle-and-data-generation/03-03-replay-stores-and-diagnostics.typ; gate: support and coverage accounting.

Evidence

- M1: docs/typst/thesis/development/m1-contract-report.typ, focused data/geometry/rollout tests, .configs/offline_only.toml, immutable store/Rerun receipts.

- M2–M4: `aria_nbv/tests/vin/`, `aria_nbv/tests/oracle/`, `aria_nbv/tests/lightning/`, and `docs/typst/thesis/sections/05-experimental-design/` plus saved calibration/ranking artifacts.
- M5–M7: rollout manifests, reports, configs, result tables, and exact support/coverage gaps in those artifacts.
- M8: CI, `make qmd-frontmatter-check`, `make thesis-pdf`, final smoke receipts, authenticated registration records, and the HM/FK07 pre-submission checklist; tracked PDF remains a development artifact and cannot establish submission.

Risks

Active risk state is pointer-only: `frame/depth aria_nbv/tests/rendering/test_depth_backprojection_conventions.py`, `target validity aria_nbv/tests/oracle/test_target_selection.py`, `rollout support aria_nbv/tests/rollouts/test_zarr_store.py`. Open blockers remain in `docs/typst/thesis/development/m1-contract-report.typ` and `.agents/issues.toml/.agents/todos.toml`; no duplicate risk narrative is maintained here.

Freeze

Freeze gate: clean reproducible smoke matrix, immutable run/config links, final coverage statement, and no stale paths or placeholders. The exact submission evidence gate remains owned by `docs/typst/thesis/main.typ` and its report bundle; this development projection does not satisfy it.

HM/FK07 release checks are human-owned and must be completed against the author’s authenticated records and the final PDF at the applicable pre- or post-submission stage:

- **Applicable rules:** confirm the individually assigned programme and SPO version in PRIMUSS; public reading versions establish only the current baseline.
- **Registration:** retain the PRIMUSS confirmation, official start date, individual deadline, assigned examiners, supervisor-specific length and form, and the attachments requested for this thesis.

- **Pre-submission readiness:** verify the final upload, mandatory attachments, and exact PDF prepared for PRIMUSS upload before the individual deadline. A repository field, local build, PDF digest, or upload alone is not submission evidence.
- **Copies:** provide a paper copy only when an examiner requests one under the current FK07 process.
- **Local final-PDF review:** as a repository/author release check, verify the declaration and AI/tool disclosure in the exact PDF prepared for PRIMUSS upload. Confirm any PRIMUSS-marked mandatory declaration, signature, or attachment against the authenticated record and assigned supervisor; this checklist does not impose a universal signature form.
- **Post-freeze submission closure:** at the actual submission, invoke *Abschlussarbeit abgeben*, retain the authenticated timestamp, and archive the exact submitted PDF. This later receipt closes the submission evidence gate but does not block the September M8 development freeze.

Official baseline checked 2026-08-25: HM ASPO, Master Informatik SPO, FK07 thesis process, PRIMUSS registration guide, and PRIMUSS submission guide. Recheck the official versions and authenticated instructions at freeze time.

Promotion queue

Promotion queue — candidate: Promote M5 lookahead/Q_H results after paired held-out evidence is immutable. Source: docs/typst/thesis/sections/05-experimental-design/; aria_nbv/tests/rollouts/test_reporting.py target: docs/typst/thesis/sections/06-results.typ and docs/typst/thesis/sections/07-discussion.typ gate: positive headroom, oracle re-scoring, uncertainty, and coverage

Promotion queue — blocked: Resolve the scale fallback before claiming population coverage. Source: docs/typst/thesis/development/m1-contract-report.typ; aria_nbv/tests/data_handling/; aria_nbv/tests/rollouts/ target: docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ gate: throughput and scene-level held-out coverage

Promotion queue — deferred: Keep online discrete Q_H and continuous target-then-pose work as bridge scope until offline gates close. Source: docs/typst/thesis/sections/07-discussion.typ; .agents/issues.toml target: docs/typst/thesis/sections/07-discussion.typ gate: stable offline Q_H and explicit scope decision

Promotion queue — rejected: Do not promote unsupported continuous, VLM, semantic-global, or real-device claims into the thesis core.

Source: docs/typst/thesis/main.typ; docs/typst/thesis/sections/07-discussion.typ target: docs/typst/thesis/sections/08-conclusion.typ gate: bounded non-claim retained

Priorities

Gate order is M1 correctness, M2–M4 target/scorer readiness, M5 headroom, M6 bridge decision, then M7/M8 freeze. Each transition requires the evidence pointer and exit gate above; no scale or bridge work bypasses a blocked prerequisite.

Issues and blockers

- M1 store, scene split, Rerun, and throughput: docs/typst/thesis/development/m1-contract-report.typ#m1-blockers.
- M5 support and headroom: docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ, docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ.
- Actionable follow-up: .agents/issues.toml and .agents/todos.toml; these records do not replace canonical owners.

M1 Contract Report

Status

Implementation: partial **Evidence:** pending

Synthetic contract guards are available, but the configured local store and scale evidence do not yet close the M1 exit gate.

Promotion gate: fresh store, scene-split, Rerun, and throughput evidence

Development source: thesis M1 gate notes and executable package owners (not a second contract owner)

Area	Status	Owner and evidence pointer
Source-contract guards	validated	Focused tests under <code>aria_nbv/tests/data_handling/</code> , <code>aria_nbv/tests/pose_generation/</code> , and <code>aria_nbv/tests/rollouts/</code> own the executable checks; this view records only their M1 status.
Configured local store	blocked	<code>.configs/offline_only.toml</code> and the resolved <code>.data/offline_cache/vin_offline</code> artifact are the smoke-evidence owners.
Scene-level split evidence	blocked	The store artifact and its split metadata require a scene-level train/validation proof before promotion.
Rerun diagnostics	blocked	No M1 Rerun recordings are present in this worktree. The recording gate remains open; absence is reported explicitly rather than linked to a nonexistent artifact.
Oracle throughput	blocked	Representative mesh, rendering, backprojection, and RRI timing remains to be captured in the experiment evidence owner.

Evidence

The current evidence view is intentionally pointer-only. The defining Python modules, tests, active configuration, and generated artifacts remain authoritative;

no schema, tensor shape, formula, serialized value, or command ledger is reproduced here.

- Synthetic guard results: `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, and `aria_nbv/tests/rollouts/`.
- Local smoke configuration and artifact root: `.configs/offline_only.toml` and `.data/offline_cache/vin_offline/`.
- Rerun evidence: absent in the current worktree; the normal/boundary/failure recording gate remains open.

Blockers

Promotion queue — blocked: Refresh the configured-store evidence and report coverage status.

Source: M1 gate notes and executable package owners target: M1 status and evidence gate: fresh immutable-store inspection

Promotion queue — blocked: Replace the one-scene diagnostic split with scene-level evidence or record the exact residual blocker.

Source: `.data/offline_cache/vin_offline/` target: M1 evidence gate: scene-level split proof

Promotion queue — blocked: Collect normal, boundary, and failure Rerun recordings.

Source: `.artifacts/rerun/` target: M1 evidence gate: three diagnostic recordings

Promotion queue — blocked: Measure representative oracle throughput and preserve the measurement artifact for review.

Source: `aria_nbv/` and active experiment configuration target: M1 evidence gate: representative timing receipt

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A Reproducibility Record

The report bundle is the single numerical interface between validated rollout artifacts and this manuscript. Its schema version, evidence status, store manifests, typed parameter rows, failure records, and sidecar hashes preserve provenance without duplicating analysis logic in Typst. The document loader rejects schema or status mismatches before any table is rendered. Compact labels and digest prefixes are used below; complete store identifiers, paths, and hashes remain in the machine-readable bundle.

Evidence class	Store	Manifest digest	Validated
Development fixture	store S1	not-a-real-m...	yes

Table 14: Loaded development-fixture provenance. The fixture verifies the document interface only and is not scientific evidence.

The bundled development fixture contains synthetic values solely to test typed loading and missingness. Its numerical rows are deliberately not rendered or interpreted.

B Development Diary

Archived source note: This marked diary is a development-only record. It is excluded from submission mode; only evidence-backed definitions, protocols, and results graduate into the thesis body.
Source: thesis mode contract

B.1 Decisions Retained

The thesis keeps the privileged oracle and actor-visible policy as separate contracts, evaluates finite candidate sets under a shared validity mask, and treats target-specific reconstruction improvement as the common comparison signal. Rollout stores retain selected transitions and provenance rather than becoming a second implementation of the oracle or training pipeline.

Open decision: Freeze the scene-disjoint analysis population and the minimum evidence scale only when the resolved manifests and exclusion ledger are available.
Source: experimental-design contract
Gate: confirmatory bundle freeze

B.2 Failures That Changed the Plan

Early rollout attempts reached the mesh-rendering path but exceeded available accelerator memory when candidate views were rendered without a bounded batch. This failure narrowed the immediate claim to implementation readiness and made peak memory, throughput, failure provenance, and completed-store validation explicit scale gates.

Validation TODO: Require completed stores, matching manifests, resource summaries, and recorded failures before extrapolating rollout bandwidth or storage demand.
Source: rollout attempt artifacts
Gate: cluster-readiness evidence

B.3 Rejected Scope

Continuous actions, online simulators, additional scene representations, and alternative reinforcement-learning families are excluded from the core study while the finite-candidate target-conditioned comparison remains unvalidated. They would

change both the action contract and the supervision regime, so adding them now would weaken rather than extend the primary experiment.

Remove or rewrite before submission: Delete bridge material that cannot be tied to a concrete limitation or a matched follow-up experiment after the confirmatory analysis is complete.

Source: scope review

Gate: discussion freeze

B.4 Directional-Memory Hypothesis

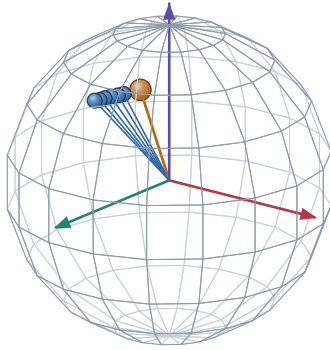
Research TODO: Treat target-centred directional memory as an ablation hypothesis, not an actor feature or contribution. The representation must be implemented, mask-audited, and compared against pose distance and overlap before any predictive claim enters the manuscript.

Source: RQ3 representation hypothesis

Gate: paired held-out ablation

A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

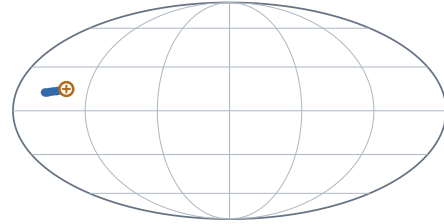


● logged history direction ● example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



$+x_e$ at map centre · $+z_e$ north · wrap seam at $\pm 180^\circ$

$$\mathbf{M}_{\text{dir}(v)} = \sum_{k < t} w_{k(v)} \mathbf{d}_{k(v)} \mathbf{d}_{k(v)}^\top$$

$$\nu_{i(v)} = 1 - \frac{\mathbf{d}_i^\top \mathbf{M}_{\text{dir}} \mathbf{d}_i}{\text{tr} \mathbf{M}_{\text{dir}} + \epsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\mathbf{M}_{\text{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.

Figure 7: Development-only directional-memory fixture. The sphere and Mollweide panels show the same logged ASE camera directions in a target-object frame; the orange query is another logged pose used to inspect one prospective second-moment novelty definition. No smooth field, spherical-harmonic representation, or learned benefit is claimed.

B.5 Next Evidence Gates

The next admissible evidence is a validated report bundle that fixes the study population, records target and candidate exclusions, supplies paired endpoint outcomes with uncertainty, and joins runtime and storage statistics to the same manifests. Only then can the manuscript replace the present availability statements with policy comparisons.

Implementation TODO: Export the confirmatory report bundle, compile both thesis modes against it, and inspect the resulting tables and failure cases before freezing claims.

Source: reporting contract

Gate: claim freeze

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen oder Hilfsmittel benutzt sowie wörtliche oder sinngemäße Zitate als solche gekennzeichnet habe.

Muenchen, 7. Jan 2027

Jan Duchscherer